

ISC Class XII Mathematics

Chapter 8 Applications of Integrals

64 Solved Problems – Numericals & Proofs

Step-by-Step ISC Board Solutions

Topics Covered

- I. Area Under a Single Curve
- II. Area Between Two Curves (Intersection-Bounded)
- III. Area of Circular/Elliptical Regions
- IV. Area of a Triangle via Integration
- V. Area of Piecewise-Defined or Sign-Changing Regions

Compiled Answer Key – Step-by-step ISC board-style solutions

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(ii)	Find the area of the region bounded by $y^2 = 9x$, $x = 2$, $x = 4$ and the x -axis in ...	1a	Numerical	8
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Q. No.	Topic / Statement Summary	Type No.	Kind	Section
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Q. No.	Topic / Statement Summary	Type No.	Kind	Section
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Q. No.	Topic / Statement Summary	Type No.	Kind	Section
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Formula & Standard Results Sheet

I. Area Under a Single Curve

Area Under a Curve (Vertical Strips)

$$\text{Area} = \int_a^b y \, dx = \int_a^b f(x) \, dx$$

Used when the curve is expressed as $y = f(x)$ and the strip is vertical, bounded above by the curve and below by the x -axis

Area Under a Curve (Horizontal Strips)

$$\text{Area} = \int_c^d x \, dy$$

Used when it is more convenient to express x as a function of y , with the strip horizontal and bounded by the y -axis

Area Symmetric About an Axis

$$\text{Area} = 2 \int_0^a y \, dx$$

For a curve symmetric about the x -axis (or y -axis), compute the area of one half and double it

Piecewise Area (Split at a Boundary)

$$\text{Area} = \int_{x_1}^{x_2} f(x) \, dx + \int_{x_2}^{x_3} g(x) \, dx$$

Used when the upper (or lower) boundary curve changes definition partway across the region

II. Area Between Two Curves

Area Between Two Curves (Vertical Strips)

$$\text{Area} = \int_a^b (y_{\text{upper}} - y_{\text{lower}}) \, dx$$

Always identify which curve lies above the other across the interval before integrating; the bounds a, b are the x -coordinates of the points of intersection

Area Between Two Curves (Horizontal Strips)

$$\text{Area} = \int_{y_1}^{y_2} (x_{\text{right}} - x_{\text{left}}) dy$$

The horizontal-strip analogue, useful when the curves are more naturally expressed as $x = g(y)$

Finding Points of Intersection

Solve the two curve equations simultaneously to find the limits of integration

Always the first step before setting up the area integral for two curves

Equation of a Line Through Two Points

$$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1} (x - x_1)$$

Needed whenever a boundary is a straight line determined by two given points, e.g. a triangle's side

III. Area of Circular and Elliptical Regions**Standard Integral for a Circular/Elliptical Arc**

$$\int \sqrt{a^2 - x^2} dx = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \left(\frac{x}{a} \right) + C$$

The core standard result behind almost every circle/ellipse area problem

Area of a Full Ellipse

$$\text{Area} = \pi ab$$

a, b are the semi-major and semi-minor axes; use the standard integral above with symmetry to derive it

Area of a Quarter Circle

$$\int_0^a \sqrt{a^2 - x^2} dx = \frac{\pi a^2}{4}$$

A direct consequence of the standard arc integral evaluated over a full quarter

Area of a Circular Sector

$$\text{Area of sector} = \frac{1}{2} R^2 \theta$$

θ in radians; often combined with an integral for the remaining segment of a partitioned circular region

Area of Overlapping/Intersecting Circles

Split the region at the point(s) of intersection and integrate each circle's area

Requires finding the intersection points of the two circles first, then setting up piecewise bounds

IV. Area of a Triangle via Integration**Triangle Area via Integration**

Find the equation of each side using two given vertices, then integrate the

Equivalent to the coordinate-geometry area formula, but derived here using definite integrals as required by the method

V. Piecewise-Defined and Sign-Changing Regions**Splitting an Absolute Value Function**

$$|f(x)| = \begin{cases} f(x), & f(x) \geq 0 \\ -f(x), & f(x) < 0 \end{cases}$$

Always split the integral at the point(s) where the expression inside the modulus changes sign

Area When the Curve Crosses the x -axis

$$\text{Area} = \int_{x_1}^{x_2} |y| dx$$

If y is negative over part of the interval, that portion contributes negatively to a plain integral – take the modulus, or split and add absolute values of each piece, to get the true geometric area

Ratio of Two Bounded Areas

Compute each area separately using the appropriate integral, then form the

Common in "find $A_1 : A_2$ " style questions – keep both computations symbolic until the final ratio step

8 – Applications of Integrals: Area Under Curves and Between Curves

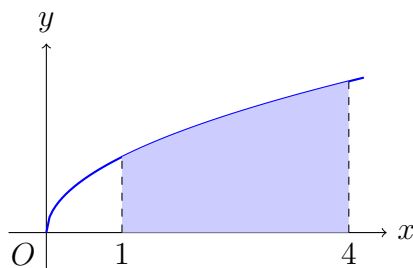
Question 1 (i)

Find the area of the region bounded by $y^2 = x$, and the lines $x = 1$, $x = 4$ and the x -axis (in the first quadrant).

TYPE 1a Area Under a Single Curve – Parabola Vertical Strip Area

Given: Equation of curve $y^2 = x$, lines $x = 1$, $x = 4$, and x -axis in first quadrant.

To Find: Area of the bounded region.



$$\text{Area under } y = \sqrt{x} \text{ from } x = 1 \text{ to } x = 4$$

Concept / Formula Used: Area $= \int_a^b y \, dx = \int_a^b f(x) \, dx$

Solution:

In the first quadrant, $y \geq 0$, so from $y^2 = x$, we have:

$$y = \sqrt{x}$$

The required area A bounded by $x = 1$, $x = 4$, and the x -axis is:

$$\begin{aligned} A &= \int_1^4 y \, dx \\ &= \int_1^4 \sqrt{x} \, dx \\ &= \int_1^4 x^{1/2} \, dx \end{aligned}$$

Using the standard integration rule $\int x^n dx = \frac{x^{n+1}}{n+1} + C$:

$$\begin{aligned}
 A &= \left[\frac{x^{3/2}}{\frac{3}{2}} \right]_1^4 \\
 &= \frac{2}{3} [x^{3/2}]_1^4 \\
 &= \frac{2}{3} (4^{3/2} - 1^{3/2}) \\
 &= \frac{2}{3} ((2^2)^{3/2} - 1) \\
 &= \frac{2}{3} (2^3 - 1) \\
 &= \frac{2}{3} (8 - 1) \\
 &= \frac{14}{3} \text{ sq. units}
 \end{aligned}$$

Final Answer

$$\text{Area} = \frac{14}{3} \text{ sq. units}$$

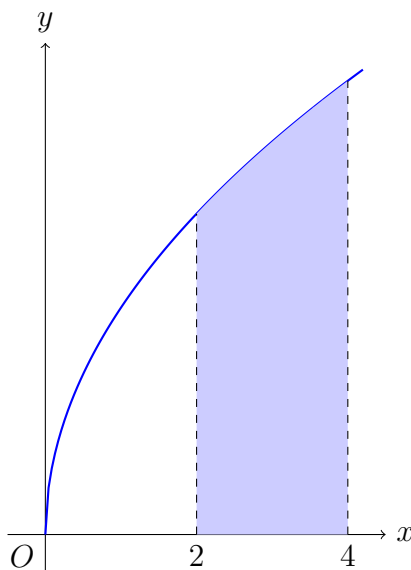
Question 1 (ii)

Find the area of the region bounded by $y^2 = 9x$, $x = 2$, $x = 4$ and the x -axis in the first quadrant.

TYPE 1a Area Under a Single Curve – Parabola Vertical Strip Area

Given: Parabola $y^2 = 9x$, lines $x = 2$, $x = 4$, x -axis, first quadrant.

To Find: Area of the bounded region.



$$\text{Area under } y = 3\sqrt{x} \text{ from } x = 2 \text{ to } x = 4$$

Concept / Formula Used: Area = $\int_a^b y \, dx$

Solution:

In the first quadrant, $y \geq 0$:

$$\begin{aligned} y^2 &= 9x \\ y &= 3\sqrt{x} \end{aligned}$$

The required area is given by:

$$\begin{aligned} A &= \int_2^4 3\sqrt{x} \, dx \\ &= 3 \int_2^4 x^{1/2} \, dx \\ &= 3 \left[\frac{2}{3} x^{3/2} \right]_2^4 \\ &= 2 (4^{3/2} - 2^{3/2}) \\ &= 2 (8 - 2\sqrt{2}) \\ &= (16 - 4\sqrt{2}) \text{ sq. units} \end{aligned}$$

Final Answer

$$\text{Area} = (16 - 4\sqrt{2}) \text{ sq. units}$$

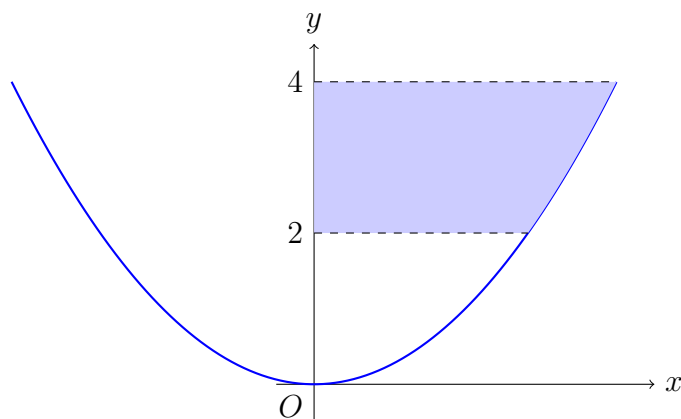
Question 2 (i)

Find the area of the region bounded by $x^2 = 4y$, $y = 2$, $y = 4$ and the y -axis in the first quadrant.

TYPE 1b Area Under a Single Curve – Horizontal Strip Integration

Given: Parabola $x^2 = 4y$, lines $y = 2$, $y = 4$, y -axis in the first quadrant.

To Find: Area of the bounded region.



Area along y -axis for parabola $x^2 = 4y$

Concept / Formula Used: Area = $\int_c^d x \, dy$

Solution:

In the first quadrant, $x \geq 0$:

$$\begin{aligned} x^2 &= 4y \\ x &= 2\sqrt{y} \end{aligned}$$

Integrating with respect to y from $y = 2$ to $y = 4$:

$$\begin{aligned} A &= \int_2^4 x \, dy \\ &= \int_2^4 2\sqrt{y} \, dy \\ &= 2 \int_2^4 y^{1/2} \, dy \\ &= 2 \left[\frac{2}{3} y^{3/2} \right]_2^4 \\ &= \frac{4}{3} (4^{3/2} - 2^{3/2}) \\ &= \frac{4}{3} (8 - 2\sqrt{2}) \\ &= \frac{32 - 8\sqrt{2}}{3} \text{ sq. units} \end{aligned}$$

Final Answer

$$\text{Area} = \frac{32 - 8\sqrt{2}}{3} \text{ sq. units}$$

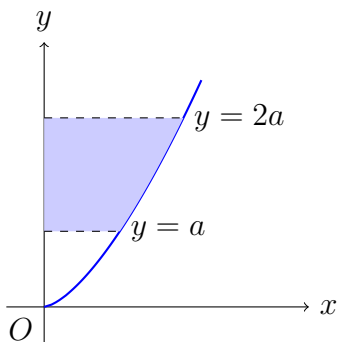
Question 2 (ii)

Find the area of the region bounded by the curve $ay^2 = x^3$, the y -axis and the lines $y = a$ and $y = 2a$.

TYPE 1b Area Under a Single Curve – Horizontal Strip Integration

Given: Curve $ay^2 = x^3$, y -axis, lines $y = a$, $y = 2a$.

To Find: Area of bounded region.



Area along y -axis for $ay^2 = x^3$

Concept / Formula Used: Area = $\int_{y_1}^{y_2} x \, dy$

Solution:

Express x in terms of y :

$$\begin{aligned}x^3 &= ay^2 \\x &= (ay^2)^{1/3} \\x &= a^{1/3} y^{2/3}\end{aligned}$$

The required area A is:

$$\begin{aligned}A &= \int_a^{2a} x \, dy \\&= \int_a^{2a} a^{1/3} y^{2/3} \, dy \\&= a^{1/3} \left[\frac{y^{5/3}}{\frac{5}{3}} \right]_a^{2a} \\&= \frac{3}{5} a^{1/3} [y^{5/3}]_a^{2a} \\&= \frac{3}{5} a^{1/3} ((2a)^{5/3} - a^{5/3}) \\&= \frac{3}{5} a^{1/3} \cdot a^{5/3} (2^{5/3} - 1) \\&= \frac{3}{5} a^2 (2^{5/3} - 1) \text{ sq. units}\end{aligned}$$

Final Answer

$$\text{Area} = \frac{3}{5} a^2 (2^{5/3} - 1) \text{ sq. units}$$

Question 3 (i)

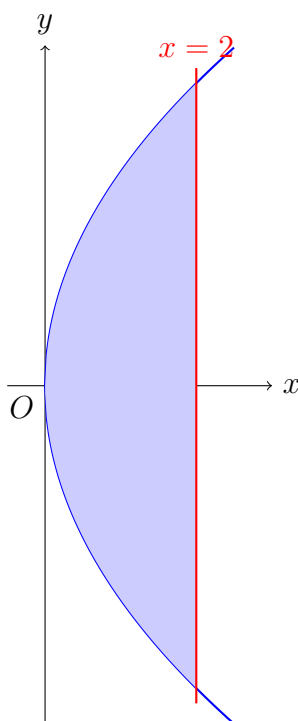
Draw a rough sketch of the curve $y^2 = 8x$ and find the area of the region bounded by the curve $y^2 = 8x$ and the line $x = 2$.

TYPE 2j Area Between Two Curves (Intersection-Bounded) – Symmetric Parabola Area

★ Likely Exam Question

Given: Parabola $y^2 = 8x$, line $x = 2$.

To Find: Rough sketch and area of the region bounded by parabola and $x = 2$.



Symmetric region bounded by $y^2 = 8x$ and $x = 2$

Concept / Formula Used: Area = $2 \int_0^a y \, dx$

Solution:

The parabola $y^2 = 8x$ is symmetric about the x -axis. For $y \geq 0$, $y = \sqrt{8x} = 2\sqrt{2}\sqrt{x}$.

Due to symmetry about the x -axis, the total area is twice the area in the first quadrant:

$$\begin{aligned}
 A &= 2 \int_0^2 y \, dx \\
 &= 2 \int_0^2 2\sqrt{2}\sqrt{x} \, dx \\
 &= 4\sqrt{2} \int_0^2 x^{1/2} \, dx \\
 &= 4\sqrt{2} \left[\frac{2}{3} x^{3/2} \right]_0^2 \\
 &= \frac{8\sqrt{2}}{3} (2^{3/2} - 0) \\
 &= \frac{8\sqrt{2}}{3} (2\sqrt{2}) \\
 &= \frac{8 \cdot 4}{3} \\
 &= \frac{32}{3} \text{ sq. units}
 \end{aligned}$$

Final Answer

$$\text{Area} = \frac{32}{3} \text{ sq. units}$$

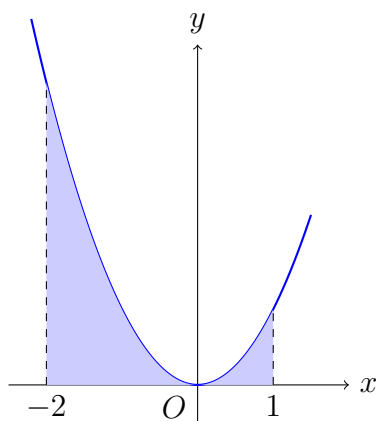
Question 3 (ii)

Using integration, find the area of the region enclosed by the curve $y = x^2$, the x -axis and the ordinates $x = -2$ and $x = 1$.

TYPE 1j Area Under a Single Curve – Standard Definite Integral Area

Given: Parabola $y = x^2$, x -axis, ordinates $x = -2$ and $x = 1$.

To Find: Area of the enclosed region.



$$\text{Area under } y = x^2 \text{ from } x = -2 \text{ to } x = 1$$

Concept / Formula Used: $\text{Area} = \int_a^b f(x) dx$

Solution:

Since $y = x^2 \geq 0$ for all $x \in [-2, 1]$, the region lies entirely above the x -axis.

$$\begin{aligned} A &= \int_{-2}^1 y dx \\ &= \int_{-2}^1 x^2 dx \\ &= \left[\frac{x^3}{3} \right]_{-2}^1 \\ &= \frac{1^3}{3} - \frac{(-2)^3}{3} \\ &= \frac{1}{3} - \left(-\frac{8}{3} \right) \\ &= \frac{1+8}{3} \\ &= \frac{9}{3} \\ &= 3 \text{ sq. units} \end{aligned}$$

Final Answer

Area = 3 sq. units

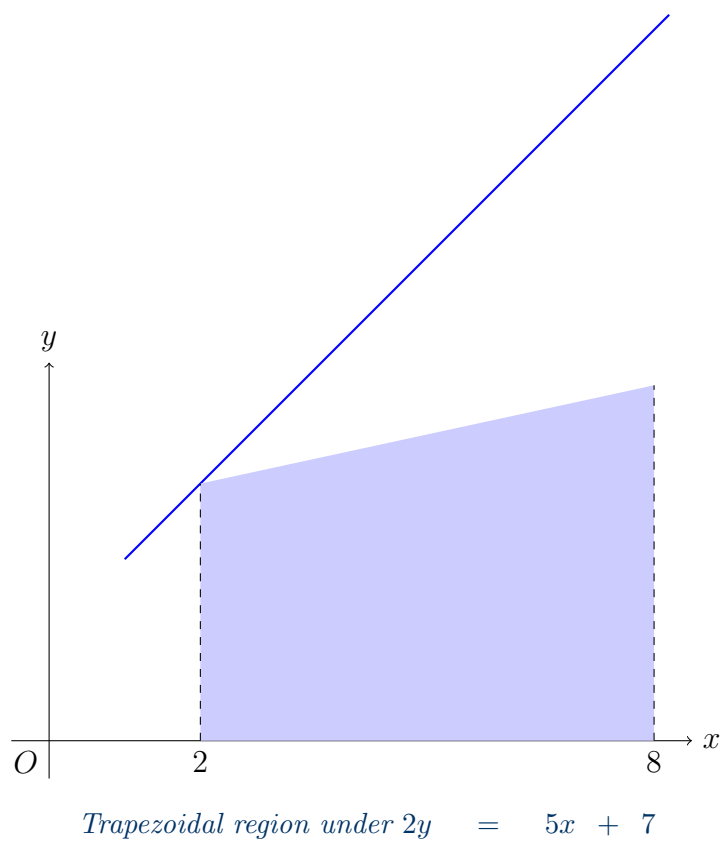
Question 4

Using integration, find the area of the region bounded by the line $2y = 5x + 7$, x -axis and the lines $x = 2$ and $x = 8$.

TYPE 1i Area Under a Single Curve – Linear Boundary Area

Given: Line $2y = 5x + 7$, x -axis, ordinate lines $x = 2$ and $x = 8$.

To Find: Area of the region.



Concept / Formula Used: Area = $\int_a^b y \, dx$

Solution:

Express y in terms of x :

$$2y = 5x + 7$$
$$y = \frac{5x + 7}{2}$$

The required area A is:

$$\begin{aligned}
 A &= \int_2^8 \left(\frac{5x+7}{2} \right) dx \\
 &= \frac{1}{2} \int_2^8 (5x+7) dx \\
 &= \frac{1}{2} \left[\frac{5x^2}{2} + 7x \right]_2^8 \\
 &= \frac{1}{2} \left(\left(\frac{5(64)}{2} + 7(8) \right) - \left(\frac{5(4)}{2} + 7(2) \right) \right) \\
 &= \frac{1}{2} ((160 + 56) - (10 + 14)) \\
 &= \frac{1}{2} (216 - 24) \\
 &= \frac{1}{2} (192) \\
 &= 96 \text{ sq. units}
 \end{aligned}$$

Final Answer

Area = 96 sq. units

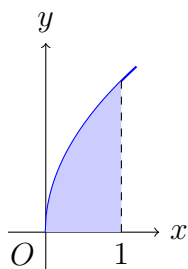
Question 5

Calculate the area under the curve $y = 2\sqrt{x}$ included between the lines $x = 0$ and $x = 1$.

TYPE 1j Area Under a Single Curve – Standard Definite Integral Area

Given: Curve $y = 2\sqrt{x}$, lines $x = 0$ and $x = 1$.

To Find: Area under the curve.



$$\text{Area under } y = 2\sqrt{x} \text{ from } x = 0 \text{ to } x = 1$$

Concept / Formula Used: Area = $\int_0^1 y dx$

Solution:

Setting up the definite integral:

$$\begin{aligned}
 A &= \int_0^1 2\sqrt{x} \, dx \\
 &= 2 \int_0^1 x^{1/2} \, dx \\
 &= 2 \left[\frac{x^{3/2}}{\frac{3}{2}} \right]_0^1 \\
 &= 2 \cdot \frac{2}{3} [x^{3/2}]_0^1 \\
 &= \frac{4}{3} (1^{3/2} - 0) \\
 &= \frac{4}{3} \text{ sq. units}
 \end{aligned}$$

Final Answer

$$\text{Area} = \frac{4}{3} \text{ sq. units}$$

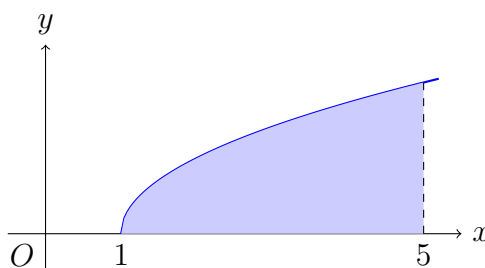
Question 6

Draw a rough sketch of the curve $y = \sqrt{x-1}$ in the interval $[1, 5]$. Find the area under the curve and between the lines $x = 1$ and $x = 5$.

TYPE 1d Area Under a Single Curve – Shifted Radical Integration

Given: Curve $y = \sqrt{x-1}$, interval $x \in [1, 5]$.

To Find: Rough sketch and area under the curve.



$$\text{Area under } y = \sqrt{x-1} \text{ from } x = 1 \text{ to } x = 5$$

Concept / Formula Used: $\text{Area} = \int_a^b \sqrt{x-a} \, dx = \left[\frac{2}{3}(x-a)^{3/2} \right]_a^b$

Solution:

The function $y = \sqrt{x-1}$ is defined for $x \geq 1$. At $x = 1$, $y = 0$; at $x = 5$, $y = \sqrt{4} = 2$.

The area A is:

$$\begin{aligned}
 A &= \int_1^5 \sqrt{x-1} \, dx \\
 &= \left[\frac{2}{3} (x-1)^{3/2} \right]_1^5 \\
 &= \frac{2}{3} ((5-1)^{3/2} - (1-1)^{3/2}) \\
 &= \frac{2}{3} (4^{3/2} - 0) \\
 &= \frac{2}{3} (8) \\
 &= \frac{16}{3} \text{ sq. units}
 \end{aligned}$$

Final Answer

$$\text{Area} = \frac{16}{3} \text{ sq. units}$$

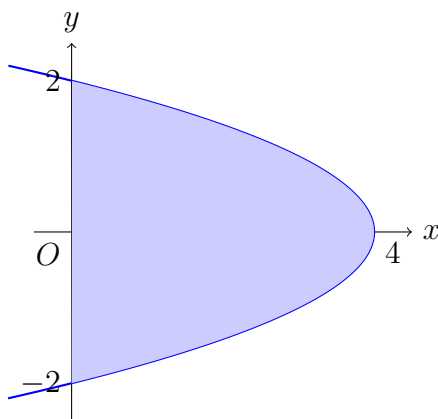
Question 7

Find the area bounded by the parabola $x = 4 - y^2$ and the y -axis.

TYPE 1h Area Under a Single Curve – Horizontal Parabola Area

Given: Parabola $x = 4 - y^2$, y -axis.

To Find: Area bounded by the curve and the y -axis.



Area enclosed by $x = 4 - y^2$ and y -axis

Concept / Formula Used: Area $= \int_c^d x \, dy$

Solution:

Find the points of intersection with the y -axis (where $x = 0$):

$$\begin{aligned}4 - y^2 &= 0 \\ y^2 &= 4 \\ y &= \pm 2\end{aligned}$$

Since the parabola is symmetric about the x -axis, the area is:

$$\begin{aligned}A &= \int_{-2}^2 (4 - y^2) dy \\ &= 2 \int_0^2 (4 - y^2) dy \\ &= 2 \left[4y - \frac{y^3}{3} \right]_0^2 \\ &= 2 \left(4(2) - \frac{2^3}{3} \right) \\ &= 2 \left(8 - \frac{8}{3} \right) \\ &= 2 \left(\frac{16}{3} \right) \\ &= \frac{32}{3} \text{ sq. units}\end{aligned}$$

Final Answer

$$\text{Area} = \frac{32}{3} \text{ sq. units}$$

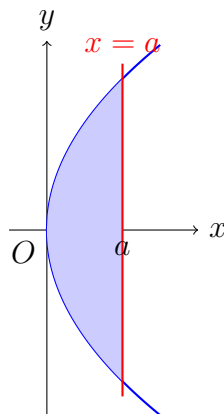
Question 8

Find the area of the region bounded by the parabola $y^2 = 4ax$ and its latus-rectum.

TYPE 2j Area Between Two Curves (Intersection-Bounded) – Symmetric Parabola Area

Given: Parabola $y^2 = 4ax$, latus-rectum line $x = a$.

To Find: Area bounded by parabola and latus-rectum.



Parabola $y^2 = 4ax$ bounded by latus-rectum $x = a$

Concept / Formula Used: Area $= 2 \int_0^a \sqrt{4ax} \, dx$

Solution:

The latus-rectum of the parabola $y^2 = 4ax$ is the line $x = a$. By symmetry about the x -axis, total area A is:

$$\begin{aligned}
 A &= 2 \int_0^a y \, dx \\
 &= 2 \int_0^a 2\sqrt{a}\sqrt{x} \, dx \\
 &= 4\sqrt{a} \int_0^a x^{1/2} \, dx \\
 &= 4\sqrt{a} \left[\frac{2}{3} x^{3/2} \right]_0^a \\
 &= \frac{8\sqrt{a}}{3} (a^{3/2}) \\
 &= \frac{8}{3} a^{1/2} \cdot a^{3/2} \\
 &= \frac{8}{3} a^2 \text{ sq. units}
 \end{aligned}$$

Final Answer

$$\text{Area} = \frac{8}{3} a^2 \text{ sq. units}$$

Question 9 (i)

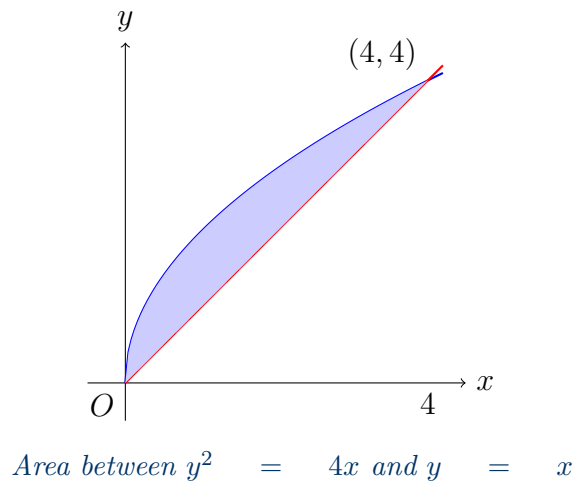
Find the area lying between the curve $y^2 = 4x$ and the line $y = x$.

TYPE 2a Area Between Two Curves (Intersection-Bounded) – Area Between Curves

★ Likely Exam Question

Given: Parabola $y^2 = 4x$, line $y = x$.

To Find: Area enclosed between parabola and line.



Concept / Formula Used: Area = $\int_a^b (y_{\text{upper}} - y_{\text{lower}}) dx$

Solution:

Find points of intersection by equating expressions for x or y :

$$\begin{aligned} y^2 &= 4x \\ x^2 &= 4x \quad (\text{substituting } y = x) \\ x^2 - 4x &= 0 \\ x(x - 4) &= 0 \implies x = 0 \quad \text{or} \quad x = 4 \end{aligned}$$

For $x \in [0, 4]$, $y_{\text{upper}} = 2\sqrt{x}$ and $y_{\text{lower}} = x$.

$$\begin{aligned} A &= \int_0^4 (2\sqrt{x} - x) dx \\ &= \left[2 \cdot \frac{2}{3} x^{3/2} - \frac{x^2}{2} \right]_0^4 \\ &= \left[\frac{4}{3} x^{3/2} - \frac{x^2}{2} \right]_0^4 \\ &= \left(\frac{4}{3} (4^{3/2}) - \frac{4^2}{2} \right) - 0 \\ &= \left(\frac{4}{3} (8) - \frac{16}{2} \right) \\ &= \frac{32}{3} - 8 \\ &= \frac{32 - 24}{3} \\ &= \frac{8}{3} \text{ sq. units} \end{aligned}$$

Final Answer

$$\text{Area} = \frac{8}{3} \text{ sq. units}$$

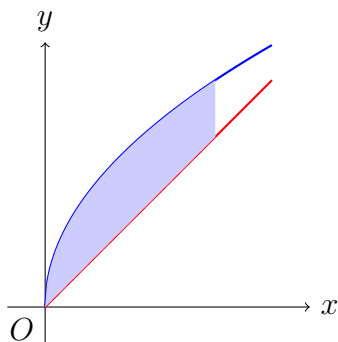
Question 9 (ii)

Find the area of the region enclosed by the parabola $y^2 = 4ax$ and the chord $y = mx$.

TYPE 2a Area Between Two Curves (Intersection-Bounded) – Area Between Curves

Given: Parabola $y^2 = 4ax$, chord line $y = mx$.

To Find: Area of enclosed region.



Area enclosed by $y^2 = 4ax$ and $y = mx$

Concept / Formula Used: $\text{Area} = \int_0^{x_1} (y_{\text{parabola}} - y_{\text{line}}) dx$

Solution:

Find points of intersection:

$$\begin{aligned} (mx)^2 &= 4ax \\ m^2x^2 - 4ax &= 0 \\ x(m^2x - 4a) &= 0 \implies x = 0 \quad \text{or} \quad x = \frac{4a}{m^2} \end{aligned}$$

Upper curve is $y = 2\sqrt{a}\sqrt{x}$, lower curve is $y = mx$:

$$\begin{aligned}
 A &= \int_0^{\frac{4a}{m^2}} (2\sqrt{a}\sqrt{x} - mx) dx \\
 &= \left[2\sqrt{a} \cdot \frac{2}{3}x^{3/2} - \frac{mx^2}{2} \right]_0^{\frac{4a}{m^2}} \\
 &= \frac{4}{3}\sqrt{a} \left(\frac{4a}{m^2} \right)^{3/2} - \frac{m}{2} \left(\frac{4a}{m^2} \right)^2 \\
 &= \frac{4}{3}\sqrt{a} \cdot \frac{8a^{3/2}}{m^3} - \frac{m}{2} \cdot \frac{16a^2}{m^4} \\
 &= \frac{32a^2}{3m^3} - \frac{8a^2}{m^3} \\
 &= \left(\frac{32 - 24}{3} \right) \frac{a^2}{m^3} \\
 &= \frac{8a^2}{3m^3} \text{ sq. units}
 \end{aligned}$$

Final Answer

$$\text{Area} = \frac{8a^2}{3m^3} \text{ sq. units}$$

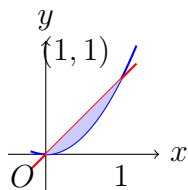
Question 10

Find the area bounded by the curves $y = x$ and $y = x^2$.

TYPE 2a Area Between Two Curves (Intersection-Bounded) – Area Between Curves

Given: Curves $y = x$ and $y = x^2$.

To Find: Area bounded between the two curves.



$$\text{Area between } y = x \text{ and } y = x^2$$

Concept / Formula Used: $\text{Area} = \int_a^b (y_{\text{upper}} - y_{\text{lower}}) dx$

Solution:

Find intersection points:

$$\begin{aligned}x^2 &= x \\x^2 - x &= 0 \\x(x - 1) &= 0 \implies x = 0, 1\end{aligned}$$

For $x \in [0, 1]$, $x \geq x^2$, so $y_{\text{upper}} = x$ and $y_{\text{lower}} = x^2$:

$$\begin{aligned}A &= \int_0^1 (x - x^2) dx \\&= \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^1 \\&= \left(\frac{1}{2} - \frac{1}{3} \right) - 0 \\&= \frac{3 - 2}{6} \\&= \frac{1}{6} \text{ sq. units}\end{aligned}$$

Final Answer

$$\text{Area} = \frac{1}{6} \text{ sq. units}$$

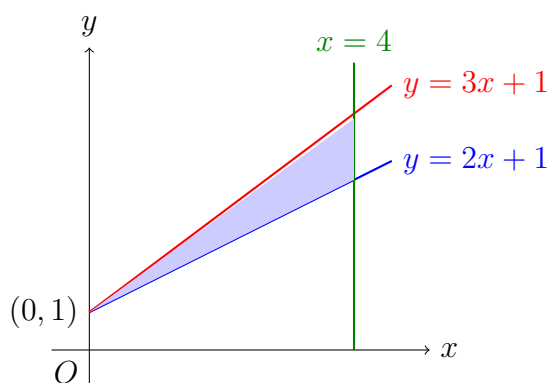
Question 11

Using definite integrals, find the area of the triangular region whose sides have the equations $y = 2x + 1$, $y = 3x + 1$ and $x = 4$.

TYPE 4b Area of a Triangle via Integration – Triangular Region Area

Given: Sides of triangle $y = 2x + 1$, $y = 3x + 1$, and $x = 4$.

To Find: Area of triangular region.



Triangular region bounded by $y = 2x + 1$, $y = 3x + 1$, $x = 4$

Concept / Formula Used: $\text{Area} = \int_a^b (y_{\text{top}} - y_{\text{bottom}}) dx$

Solution:

Find intersection of $y = 2x + 1$ and $y = 3x + 1$:

$$\begin{aligned} 2x + 1 &= 3x + 1 \\ x = 0 &\implies y = 1 \end{aligned}$$

The vertex is $(0, 1)$. The region extends from $x = 0$ to $x = 4$. For $x \in [0, 4]$, $3x+1 \geq 2x+1$.

$$\begin{aligned} A &= \int_0^4 ((3x + 1) - (2x + 1)) dx \\ &= \int_0^4 x dx \\ &= \left[\frac{x^2}{2} \right]_0^4 \\ &= \frac{4^2}{2} - 0 \\ &= \frac{16}{2} \\ &= 8 \text{ sq. units} \end{aligned}$$

Final Answer

Area = 8 sq. units

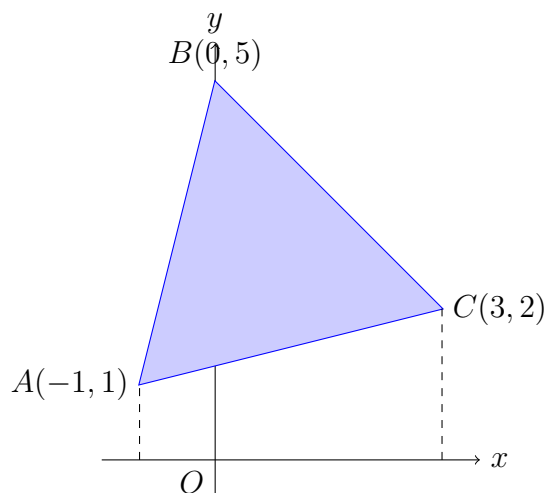
Question 12 (i)

Using integration, find the area of the triangular region whose vertices are $(-1, 1)$, $(0, 5)$ and $(3, 2)$.

TYPE 4a Area of a Triangle via Integration – Triangle Vertices Integration

Given: Vertices $A(-1, 1)$, $B(0, 5)$, $C(3, 2)$.

To Find: Area of triangle ABC .



Triangle ABC with vertices $(-1, 1)$, $(0, 5)$, $(3, 2)$

Concept / Formula Used: (x_1, y_1) (x_2, y_2) $y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$

Solution:

Find side line equations: 1. Equation of AB joining $(-1, 1)$ and $(0, 5)$:

$$y - 1 = \frac{5 - 1}{0 - (-1)}(x - (-1))$$

$$y - 1 = 4(x + 1) \implies y = 4x + 5$$

2. Equation of BC joining $(0, 5)$ and $(3, 2)$:

$$y - 5 = \frac{2 - 5}{3 - 0}(x - 0)$$

$$y - 5 = -1(x) \implies y = 5 - x$$

3. Equation of AC joining $(-1, 1)$ and $(3, 2)$:

$$y - 1 = \frac{2 - 1}{3 - (-1)}(x + 1)$$

$$y - 1 = \frac{1}{4}(x + 1) \implies y = \frac{x + 5}{4}$$

Area $A = \int_{-1}^0 (y_{AB} - y_{AC}) dx + \int_0^3 (y_{BC} - y_{AC}) dx$:

$$A_1 = \int_{-1}^0 \left((4x + 5) - \frac{x + 5}{4} \right) dx = \int_{-1}^0 \left(\frac{15x + 15}{4} \right) dx$$

$$= \frac{15}{4} \left[\frac{x^2}{2} + x \right]_{-1}^0 = \frac{15}{4} \left(0 - \left(\frac{1}{2} - 1 \right) \right) = \frac{15}{8}$$

$$A_2 = \int_0^3 \left((5 - x) - \frac{x + 5}{4} \right) dx = \int_0^3 \left(\frac{15 - 5x}{4} \right) dx$$

$$= \frac{5}{4} \left[3x - \frac{x^2}{2} \right]_0^3 = \frac{5}{4} \left(9 - \frac{9}{2} \right) = \frac{45}{8}$$

$$A = A_1 + A_2 = \frac{15}{8} + \frac{45}{8} = \frac{60}{8} = \frac{15}{2} \text{ sq. units}$$

Final Answer

$$\text{Area} = \frac{15}{2} \text{ sq. units}$$

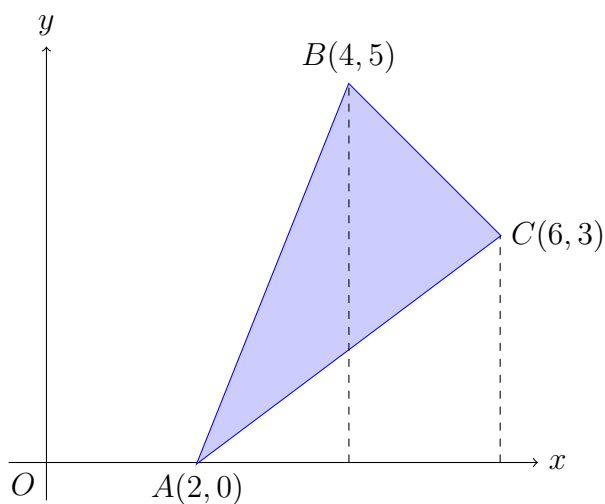
Question 12 (ii)

Using integration, find the area of the triangle ABC, the coordinates of whose vertices are $A(2, 0)$, $B(4, 5)$ and $C(6, 3)$.

TYPE 4a Area of a Triangle via Integration – Triangle Vertices Integration

Given: Vertices $A(2, 0)$, $B(4, 5)$, $C(6, 3)$.

To Find: Area of triangle ABC .



Triangle ABC with vertices $(2, 0)$, $(4, 5)$, $(6, 3)$

Concept / Formula Used: $y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$

Solution:

Find line equations: 1. AB : $y - 0 = \frac{5-0}{4-2}(x - 2) \implies y = \frac{5}{2}(x - 2)$ 2. BC : $y - 5 = \frac{3-5}{6-4}(x - 4) \implies y = 9 - x$ 3. AC : $y - 0 = \frac{3-0}{6-2}(x - 2) \implies y = \frac{3}{4}(x - 2)$

Area $A = \int_2^4 (y_{AB} - y_{AC}) dx + \int_4^6 (y_{BC} - y_{AC}) dx$:

$$\begin{aligned} A_1 &= \int_2^4 \left(\frac{5}{2}(x-2) - \frac{3}{4}(x-2) \right) dx = \frac{7}{4} \int_2^4 (x-2) dx \\ &= \frac{7}{4} \left[\frac{(x-2)^2}{2} \right]_2^4 = \frac{7}{4} (2) = \frac{7}{2} \\ A_2 &= \int_4^6 \left((9-x) - \frac{3}{4}(x-2) \right) dx = \int_4^6 \left(\frac{21}{2} - \frac{7x}{4} \right) dx \\ &= \left[\frac{21x}{2} - \frac{7x^2}{8} \right]_4^6 = \left(63 - \frac{63}{2} \right) - (42 - 14) = \frac{63}{2} - 28 = \frac{7}{2} \\ A &= A_1 + A_2 = \frac{7}{2} + \frac{7}{2} = 7 \text{ sq. units} \end{aligned}$$

Final Answer

Area = 7 sq. units

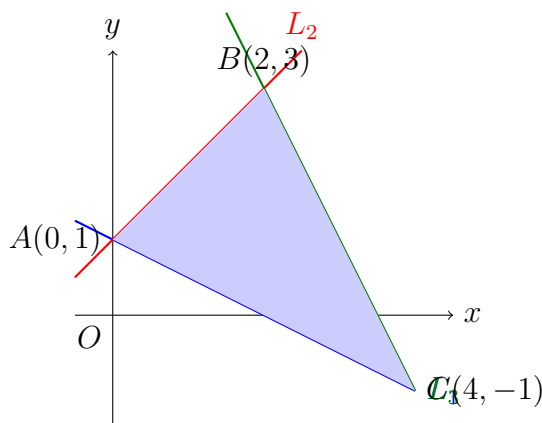
Question 13 (i)

Using integration, find the area bounded by the lines $x + 2y = 2$, $y - x = 1$ and $2x + y = 7$.

TYPE 2m Area Between Two Curves (Intersection-Bounded) – Lines Intersection Area

Given: Lines $L_1 : x + 2y = 2$, $L_2 : y - x = 1$, $L_3 : 2x + y = 7$.

To Find: Area enclosed by the three lines.



Region bounded by three intersecting lines

Concept / Formula Used: Area = $\int_{x_1}^{x_2} (y_{\text{top1}} - y_{\text{bot}}) dx + \int_{x_2}^{x_3} (y_{\text{top2}} - y_{\text{bot}}) dx$

Solution:

Find the vertices by solving pairs of equations: 1. $L_1 \cap L_2$: $x + 2(x + 1) = 2 \implies 3x = 0 \implies x = 0, y = 1$. Point $A(0, 1)$. 2. $L_2 \cap L_3$: $2x + (x + 1) = 7 \implies 3x = 6 \implies x =$

2, $y = 3$. Point $B(2, 3)$. 3. $L_1 \cap L_3: x = 2 - 2y \implies 2(2 - 2y) + y = 7 \implies -3y = 3 \implies y = -1, x = 4$. Point $C(4, -1)$.

Split integration into two regions: $[0, 2]$ and $[2, 4]$. Bottom boundary is $L_1 \implies y = \frac{2-x}{2}$.

$$A_1 = \int_0^2 \left((x+1) - \frac{2-x}{2} \right) dx = \int_0^2 \frac{3x}{2} dx = \left[\frac{3x^2}{4} \right]_0^2 = 3$$

$$A_2 = \int_2^4 \left((7-2x) - \frac{2-x}{2} \right) dx = \int_2^4 \left(6 - \frac{3x}{2} \right) dx = \left[6x - \frac{3x^2}{4} \right]_2^4$$

$$= (24 - 12) - (12 - 3) = 12 - 9 = 3$$

$$A = A_1 + A_2 = 3 + 3 = 6 \text{ sq. units}$$

Final Answer

Area = 6 sq. units

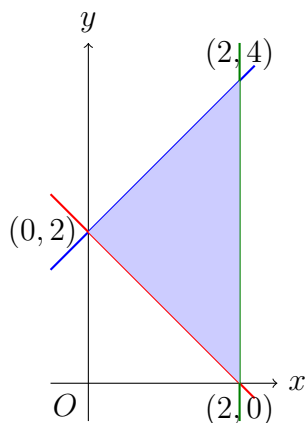
Question 13 (ii)

Using integration, find the area of the region bounded by the lines $y = 2 + x$, $y = 2 - x$ and $x = 2$.

TYPE 2m Area Between Two Curves (Intersection-Bounded) – Lines Intersection Area

Given: Lines $y = 2 + x$, $y = 2 - x$, $x = 2$.

To Find: Area of bounded region.



Triangular region formed by $y = 2 + x$, $y = 2 - x$, $x = 2$

Concept / Formula Used: Area = $\int_0^2 (y_{\text{upper}} - y_{\text{lower}}) dx$

Solution:

Intersection of $y = 2 + x$ and $y = 2 - x$:

$$2 + x = 2 - x \implies 2x = 0 \implies x = 0, y = 2$$

For $x \in [0, 2]$, upper line is $y = 2 + x$ and lower line is $y = 2 - x$:

$$\begin{aligned}
 A &= \int_0^2 ((2 + x) - (2 - x)) dx \\
 &= \int_0^2 2x dx \\
 &= [x^2]_0^2 \\
 &= 2^2 - 0 \\
 &= 4 \text{ sq. units}
 \end{aligned}$$

Final Answer

Area = 4 sq. units

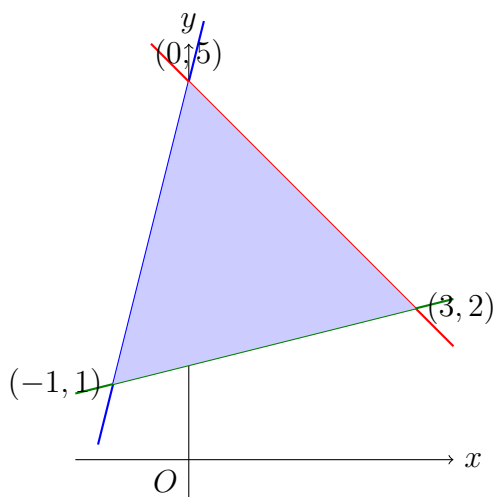
Question 13 (iii)

Using integration, find the area of the region bounded by the lines $4x - y + 5 = 0$, $x + y - 5 = 0$ and $x - 4y + 5 = 0$.

TYPE 2m Area Between Two Curves (Intersection-Bounded) – Lines Intersection Area

Given: Lines $L_1 : 4x - y + 5 = 0$, $L_2 : x + y - 5 = 0$, $L_3 : x - 4y + 5 = 0$.

To Find: Area bounded by the lines.



Triangle formed by L_1, L_2, L_3

Concept / Formula Used: Area = $\int_{x_1}^{x_2} (y_{\text{top1}} - y_{\text{bot}}) dx + \int_{x_2}^{x_3} (y_{\text{top2}} - y_{\text{bot}}) dx$

Solution:

Find vertices: 1. $L_1 \cap L_2$: $4x + 5 = 5 - x \implies 5x = 0 \implies x = 0, y = 5$. 2. $L_1 \cap L_3$: $4x + 5 = \frac{x+5}{4} \implies 16x + 20 = x + 5 \implies 15x = -15 \implies x = -1, y = 1$. 3. $L_2 \cap L_3$: $5 - x = \frac{x+5}{4} \implies 20 - 4x = x + 5 \implies 5x = 15 \implies x = 3, y = 2$.

Bottom boundary is $L_3 \Rightarrow y = \frac{x+5}{4}$.

$$A_1 = \int_{-1}^0 \left((4x+5) - \frac{x+5}{4} \right) dx = \int_{-1}^0 \frac{15x+15}{4} dx = \frac{15}{8}$$

$$A_2 = \int_0^3 \left((5-x) - \frac{x+5}{4} \right) dx = \int_0^3 \frac{15-5x}{4} dx = \frac{45}{8}$$

$$A = A_1 + A_2 = \frac{15}{8} + \frac{45}{8} = \frac{60}{8} = \frac{15}{2} = 7.5 \text{ sq. units}$$

Final Answer

$$\text{Area} = \frac{15}{2} \text{ sq. units}$$

Question 14

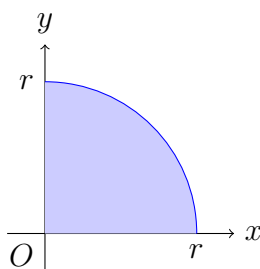
Determine the area under the curve $y = \sqrt{r^2 - x^2}$ included between the lines $x = 0$ and $x = r$. Hence, prove that the area of a circle of radius r is πr^2 .

TYPE 3c Area of Circular/Elliptical Regions – Circle Quadrant Derivation

★ Likely Exam Question

Given: Curve $y = \sqrt{r^2 - x^2}$, limits $x = 0$ to $x = r$.

To Prove: Area under curve is $\frac{\pi r^2}{4}$ and area of circle of radius r is πr^2 .



First quadrant sector of circle $x^2 + y^2 = r^2$

Concept / Formula Used: $\int \sqrt{a^2 - x^2} dx = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \left(\frac{x}{a} \right) + C$

Solution:

Let I be the area under $y = \sqrt{r^2 - x^2}$ from $x = 0$ to $x = r$:

$$\begin{aligned}
 I &= \int_0^r \sqrt{r^2 - x^2} \, dx \\
 &= \left[\frac{x}{2} \sqrt{r^2 - x^2} + \frac{r^2}{2} \sin^{-1} \left(\frac{x}{r} \right) \right]_0^r \\
 &= \left(0 + \frac{r^2}{2} \sin^{-1}(1) \right) - \left(0 + \frac{r^2}{2} \sin^{-1}(0) \right) \\
 &= \frac{r^2}{2} \left(\frac{\pi}{2} \right) - 0 \\
 &= \frac{\pi r^2}{4} \text{ sq. units}
 \end{aligned}$$

Since the circle $x^2 + y^2 = r^2$ is symmetric about both axes, it consists of four such equal quadrant regions:

$$\begin{aligned}
 \text{Total Area of Circle} &= 4 \times I \\
 &= 4 \times \frac{\pi r^2}{4} \\
 &= \pi r^2 \text{ sq. units}
 \end{aligned}$$

Hence Proved

Area of circle of radius r is πr^2 .

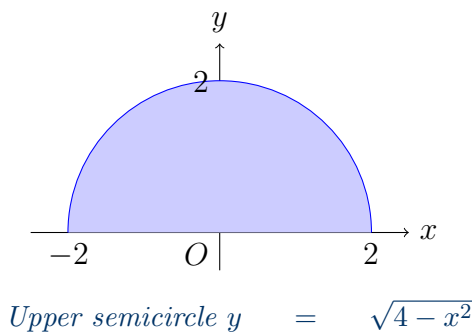
Question 15

Sketch the region $\{(x, y) : y = \sqrt{4 - x^2}\}$ and x -axis. Find the area of the region using integration.

TYPE 1c Area Under a Single Curve – Semicircle Integration

Given: Region bounded by $y = \sqrt{4 - x^2}$ and x -axis.

To Find: Sketch and area of the region.



Concept / Formula Used: $\int \sqrt{a^2 - x^2} \, dx = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \left(\frac{x}{a} \right)$

Solution:

The equation $y = \sqrt{4 - x^2}$ represents the upper half of the circle $x^2 + y^2 = 4$ of radius $r = 2$. It intersects the x -axis at $x = -2$ and $x = 2$. By symmetry:

$$\begin{aligned}
 A &= \int_{-2}^2 \sqrt{4 - x^2} \, dx \\
 &= 2 \int_0^2 \sqrt{4 - x^2} \, dx \\
 &= 2 \left[\frac{x}{2} \sqrt{4 - x^2} + \frac{4}{2} \sin^{-1} \left(\frac{x}{2} \right) \right]_0^2 \\
 &= 2 \left((0 + 2 \sin^{-1}(1)) - 0 \right) \\
 &= 2 \left(2 \cdot \frac{\pi}{2} \right) \\
 &= 2\pi \text{ sq. units}
 \end{aligned}$$

Final Answer

Area = 2π sq. units

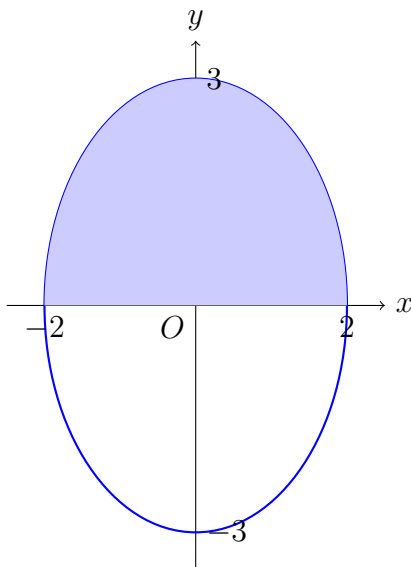
Question 16

Make a rough sketch of the curve $\frac{x^2}{4} + \frac{y^2}{9} = 1$ and find: (i) the area under the curve and above the x -axis. (ii) the area enclosed by the curve.

TYPE 3a Area of Circular/Elliptical Regions – Standard Ellipse Area

Given: Ellipse equation $\frac{x^2}{4} + \frac{y^2}{9} = 1$, so $a = 2$, $b = 3$.

To Find: Sketch, (i) area above x -axis, (ii) total area of ellipse.



$$\text{Ellipse } \frac{x^2}{4} + \frac{y^2}{9} = 1$$

Concept / Formula Used: Area of Ellipse = πab

Solution:

From $\frac{x^2}{4} + \frac{y^2}{9} = 1$, express y in terms of x :

$$\begin{aligned}\frac{y^2}{9} &= 1 - \frac{x^2}{4} \\ y^2 &= \frac{9}{4}(4 - x^2) \implies y = \frac{3}{2}\sqrt{4 - x^2} \quad \text{for } y \geq 0\end{aligned}$$

(i) Area above the x -axis:

$$\begin{aligned}A_1 &= \int_{-2}^2 \frac{3}{2}\sqrt{4 - x^2} dx \\ &= 2 \cdot \frac{3}{2} \int_0^2 \sqrt{4 - x^2} dx \\ &= 3 \left[\frac{x}{2}\sqrt{4 - x^2} + \frac{4}{2} \sin^{-1} \left(\frac{x}{2} \right) \right]_0^2 \\ &= 3 \left(2 \cdot \frac{\pi}{2} \right) = 3\pi \text{ sq. units}\end{aligned}$$

(ii) Total area enclosed by the ellipse:

$$A_{\text{total}} = 2 \times A_1 = 2 \times 3\pi = 6\pi \text{ sq. units}$$

Final Answer

(i) 3π sq. units, (ii) 6π sq. units

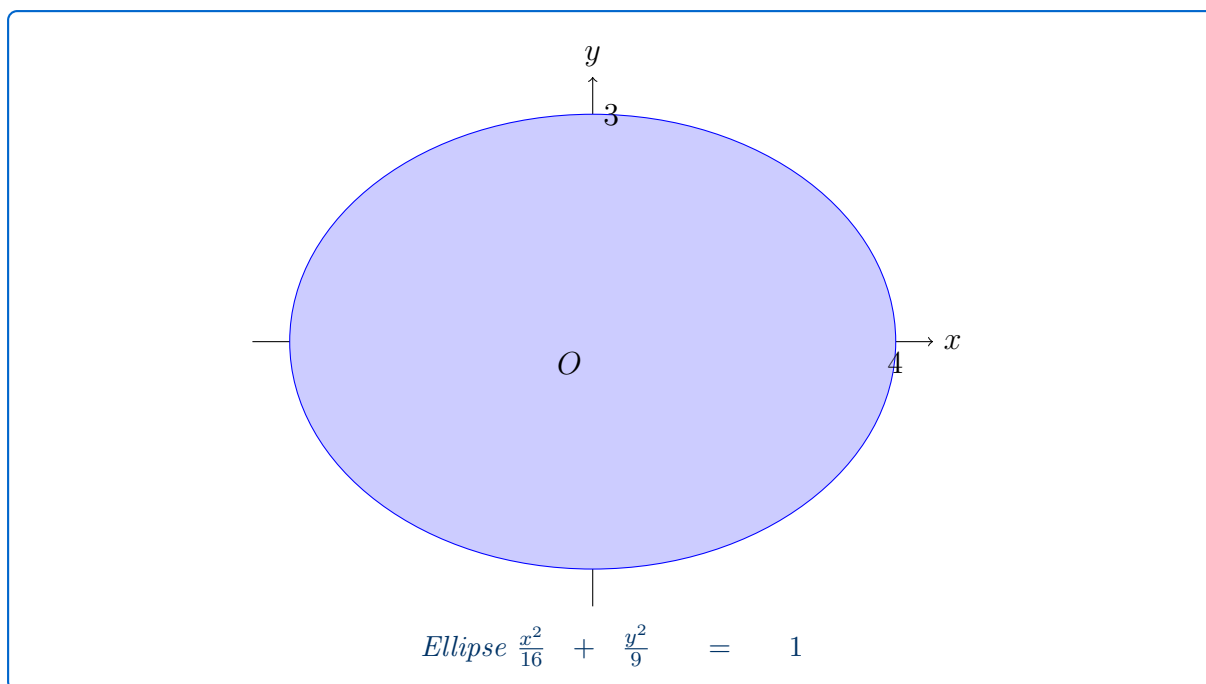
Question 17 (i)

Find the area of the region bounded by the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$.

TYPE 3a Area of Circular/Elliptical Regions – Standard Ellipse Area

Given: Ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$, where $a = 4$ and $b = 3$.

To Find: Area enclosed by ellipse.



Concept / Formula Used: Area = $4 \int_0^a y \, dx$

Solution:

Express y in terms of x :

$$y = \frac{3}{4} \sqrt{16 - x^2}$$

By four-fold symmetry:

$$\begin{aligned} A &= 4 \int_0^4 \frac{3}{4} \sqrt{16 - x^2} \, dx \\ &= 3 \int_0^4 \sqrt{16 - x^2} \, dx \\ &= 3 \left[\frac{x}{2} \sqrt{16 - x^2} + \frac{16}{2} \sin^{-1} \left(\frac{x}{4} \right) \right]_0^4 \\ &= 3 (0 + 8 \sin^{-1}(1)) \\ &= 3 \left(8 \cdot \frac{\pi}{2} \right) \\ &= 12\pi \text{ sq. units} \end{aligned}$$

Final Answer

Area = 12π sq. units

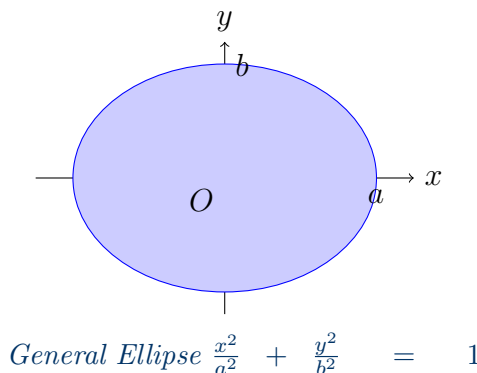
Question 17 (ii)

Find the area enclosed by the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

TYPE 3a Area of Circular/Elliptical Regions – Standard Ellipse Area

Given: Ellipse equation $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

To Prove: Area enclosed by standard ellipse is πab .



Concept / Formula Used: $\int_0^a \sqrt{a^2 - x^2} dx = \frac{\pi a^2}{4}$

Solution:

From $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$:

$$y = \frac{b}{a} \sqrt{a^2 - x^2}$$

Due to symmetry across four quadrants:

$$\begin{aligned} A &= 4 \int_0^a y dx \\ &= 4 \int_0^a \frac{b}{a} \sqrt{a^2 - x^2} dx \\ &= \frac{4b}{a} \left[\frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \left(\frac{x}{a} \right) \right]_0^a \\ &= \frac{4b}{a} \left(0 + \frac{a^2}{2} \sin^{-1}(1) \right) \\ &= \frac{4b}{a} \left(\frac{a^2}{2} \cdot \frac{\pi}{2} \right) \\ &= \pi ab \text{ sq. units} \end{aligned}$$

Hence Proved

Area enclosed by ellipse is πab sq. units.

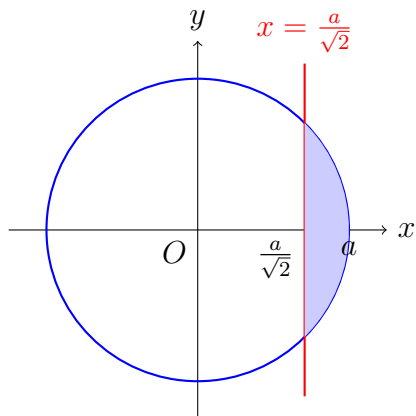
Question 18

Find the area of the minor segment of the circle $x^2 + y^2 = a^2$ cut off by the line $x = \frac{a}{\sqrt{2}}$.

TYPE 3d Area of Circular/Elliptical Regions – Circular Segment Area

Given: Circle $x^2 + y^2 = a^2$, vertical line $x = \frac{a}{\sqrt{2}}$.

To Find: Area of the minor segment.



Minor segment cut off by $x = a/\sqrt{2}$

Concept / Formula Used: Area $= 2 \int_{a/\sqrt{2}}^a \sqrt{a^2 - x^2} dx$

Solution:

By symmetry about the x -axis:

$$\begin{aligned}
 A &= 2 \int_{\frac{a}{\sqrt{2}}}^a \sqrt{a^2 - x^2} dx \\
 &= 2 \left[\frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \left(\frac{x}{a} \right) \right]_{\frac{a}{\sqrt{2}}}^a \\
 &= 2 \left[\left(0 + \frac{a^2}{2} \sin^{-1}(1) \right) - \left(\frac{a}{2\sqrt{2}} \sqrt{a^2 - \frac{a^2}{2}} + \frac{a^2}{2} \sin^{-1} \left(\frac{1}{\sqrt{2}} \right) \right) \right] \\
 &= 2 \left[\frac{\pi a^2}{4} - \left(\frac{a}{2\sqrt{2}} \cdot \frac{a}{\sqrt{2}} + \frac{a^2}{2} \cdot \frac{\pi}{4} \right) \right] \\
 &= 2 \left[\frac{\pi a^2}{4} - \frac{a^2}{4} - \frac{\pi a^2}{8} \right] \\
 &= 2 \left[\frac{\pi a^2}{8} - \frac{a^2}{4} \right] \\
 &= \frac{a^2}{2} \left(\frac{\pi}{2} - 1 \right) \text{ sq. units}
 \end{aligned}$$

Final Answer

$$\text{Area} = \frac{a^2}{2} \left(\frac{\pi}{2} - 1 \right) \text{ sq. units}$$

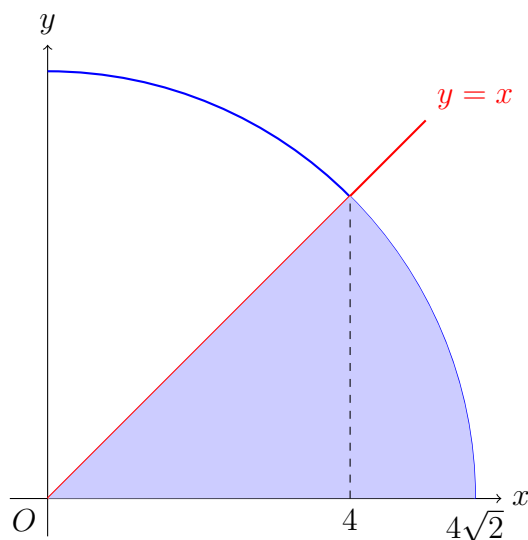
Question 19 (i)

Find the area of the region in the first quadrant enclosed by the x -axis, the line $y = x$ and the circle $x^2 + y^2 = 32$.

TYPE 3b Area of Circular/Elliptical Regions – Circle Sector Partition

Given: Circle $x^2 + y^2 = 32$, line $y = x$, x -axis in 1st quadrant.

To Find: Area of the region.



First quadrant region bounded by $y = x$ and $x^2 + y^2 = 32$

Concept / Formula Used: Area = $\int_0^{x_1} y_{\text{line}} dx + \int_{x_1}^R y_{\text{circle}} dx$

Solution:

Intersection of $y = x$ and $x^2 + y^2 = 32$:

$$x^2 + x^2 = 32 \implies 2x^2 = 32 \implies x = 4 \quad (x > 0)$$

Radius $R = \sqrt{32} = 4\sqrt{2}$. Split integral at $x = 4$:

$$\begin{aligned} A_1 &= \int_0^4 x dx = \left[\frac{x^2}{2} \right]_0^4 = 8 \\ A_2 &= \int_4^{4\sqrt{2}} \sqrt{32 - x^2} dx = \left[\frac{x}{2} \sqrt{32 - x^2} + 16 \sin^{-1} \left(\frac{x}{4\sqrt{2}} \right) \right]_4^{4\sqrt{2}} \\ &= \left(0 + 16 \sin^{-1}(1) \right) - \left(\frac{4}{2} \sqrt{16} + 16 \sin^{-1} \left(\frac{1}{\sqrt{2}} \right) \right) \\ &= 16 \left(\frac{\pi}{2} \right) - \left(8 + 16 \left(\frac{\pi}{4} \right) \right) = 8\pi - 8 - 4\pi = 4\pi - 8 \\ A &= A_1 + A_2 = 8 + (4\pi - 8) = 4\pi \text{ sq. units} \end{aligned}$$

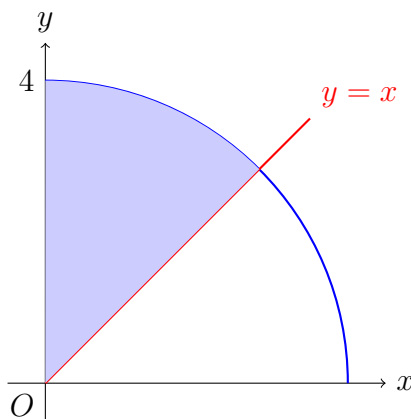
Final AnswerArea = 4π sq. units**Question 19 (ii)**

Using integration, find the area of the region bounded by the circle $x^2 + y^2 = 16$, line $y = x$ and y -axis, but lying in the 1st quadrant.

TYPE 3b Area of Circular/Elliptical Regions – Circle Sector Partition

Given: Circle $x^2 + y^2 = 16$, line $y = x$, y -axis in 1st quadrant.

To Find: Area bounded by line, y -axis, and circle.



Region between y -axis, $y = x$, and $x^2 + y^2 = 16$

Concept / Formula Used: Area = $\int_0^{y_1} x_{\text{line}} dy + \int_{y_1}^R x_{\text{circle}} dy$

Solution:

Intersection of $y = x$ and $x^2 + y^2 = 16$:

$$x^2 + x^2 = 16 \implies 2x^2 = 16 \implies x = 2\sqrt{2}, y = 2\sqrt{2}$$

Integrating with respect to y :

$$\begin{aligned} A &= \int_0^{2\sqrt{2}} y dy + \int_{2\sqrt{2}}^4 \sqrt{16 - y^2} dy \\ &= \left[\frac{y^2}{2} \right]_0^{2\sqrt{2}} + \left[\frac{y}{2} \sqrt{16 - y^2} + 8 \sin^{-1} \left(\frac{y}{4} \right) \right]_{2\sqrt{2}}^4 \\ &= 4 + (0 + 8 \sin^{-1}(1)) - \left(\sqrt{2}\sqrt{8} + 8 \sin^{-1} \left(\frac{1}{\sqrt{2}} \right) \right) \\ &= 4 + 4\pi - (4 + 2\pi) \\ &= 2\pi \text{ sq. units} \end{aligned}$$

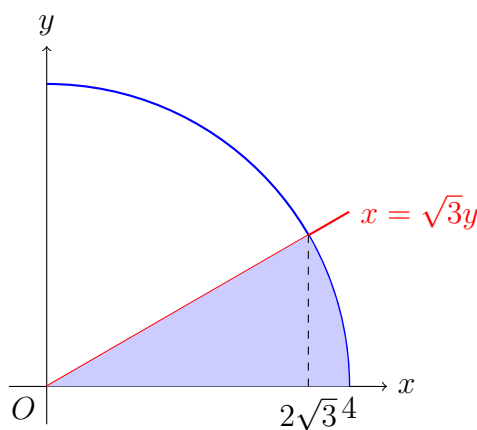
Final AnswerArea = 2π sq. units**Question 19 (iii)**

Find the area bounded by the circle $x^2 + y^2 = 16$ and the line $\sqrt{3}y = x$ in the first quadrant, by using integration.

TYPE 3b Area of Circular/Elliptical Regions – Circle Sector Partition

Given: Circle $x^2 + y^2 = 16$, line $\sqrt{3}y = x \Rightarrow y = \frac{x}{\sqrt{3}}$, x -axis, 1st quadrant.

To Find: Area of the bounded region.



Region between x -axis, $x = \sqrt{3}y$, and circle

Concept / Formula Used: Area = $\int_0^{2\sqrt{3}} \frac{x}{\sqrt{3}} dx + \int_{2\sqrt{3}}^4 \sqrt{16 - x^2} dx$

Solution:

Intersection point:

$$x^2 + \left(\frac{x}{\sqrt{3}}\right)^2 = 16 \Rightarrow \frac{4x^2}{3} = 16 \Rightarrow x^2 = 12 \Rightarrow x = 2\sqrt{3}$$

Calculate area parts:

$$\begin{aligned} A_1 &= \int_0^{2\sqrt{3}} \frac{x}{\sqrt{3}} dx = \frac{1}{\sqrt{3}} \left[\frac{x^2}{2} \right]_0^{2\sqrt{3}} = \frac{6}{\sqrt{3}} = 2\sqrt{3} \\ A_2 &= \int_{2\sqrt{3}}^4 \sqrt{16 - x^2} dx = \left[\frac{x}{2} \sqrt{16 - x^2} + 8 \sin^{-1} \left(\frac{x}{4} \right) \right]_{2\sqrt{3}}^4 \\ &= 4\pi - \left(\sqrt{3}\sqrt{4} + 8 \sin^{-1} \left(\frac{\sqrt{3}}{2} \right) \right) = 4\pi - 2\sqrt{3} - \frac{8\pi}{3} \\ A &= A_1 + A_2 = 2\sqrt{3} + 4\pi - 2\sqrt{3} - \frac{8\pi}{3} = \frac{4\pi}{3} \text{ sq. units} \end{aligned}$$

Final Answer

$$\text{Area} = \frac{4\pi}{3} \text{ sq. units}$$

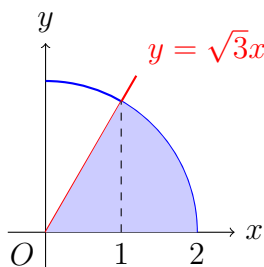
Question 19 (iv)

Using integration, find the area of the region enclosed by the line $y = \sqrt{3}x$, semi-circle $y = \sqrt{4 - x^2}$ and x -axis in the first quadrant.

TYPE 3b Area of Circular/Elliptical Regions – Circle Sector Partition

Given: Line $y = \sqrt{3}x$, semicircle $y = \sqrt{4 - x^2}$, x -axis in 1st quadrant.

To Find: Area of the region.



Region bounded by $y = \sqrt{3}x$, $y = \sqrt{4 - x^2}$, and x -axis

Concept / Formula Used: Area = $\int_0^1 \sqrt{3}x \, dx + \int_1^2 \sqrt{4 - x^2} \, dx$

Solution:

Intersection point:

$$\sqrt{3}x = \sqrt{4 - x^2} \implies 3x^2 = 4 - x^2 \implies 4x^2 = 4 \implies x = 1$$

Evaluating the area:

$$\begin{aligned} A_1 &= \int_0^1 \sqrt{3}x \, dx = \left[\frac{\sqrt{3}x^2}{2} \right]_0^1 = \frac{\sqrt{3}}{2} \\ A_2 &= \int_1^2 \sqrt{4 - x^2} \, dx = \left[\frac{x}{2} \sqrt{4 - x^2} + 2 \sin^{-1} \left(\frac{x}{2} \right) \right]_1^2 \\ &= (0 + 2 \sin^{-1}(1)) - \left(\frac{\sqrt{3}}{2} + 2 \sin^{-1} \left(\frac{1}{2} \right) \right) \\ &= \pi - \frac{\sqrt{3}}{2} - \frac{\pi}{3} = \frac{2\pi}{3} - \frac{\sqrt{3}}{2} \\ A &= A_1 + A_2 = \frac{\sqrt{3}}{2} + \frac{2\pi}{3} - \frac{\sqrt{3}}{2} = \frac{2\pi}{3} \text{ sq. units} \end{aligned}$$

Final Answer

$$\text{Area} = \frac{2\pi}{3} \text{ sq. units}$$

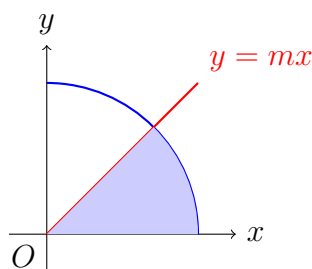
Question 20

The area of the region bounded by the lines $y = mx$ ($m > 0$), the curve $x^2 + y^2 = 4$ and the x -axis in the first quadrant is $\frac{\pi}{2}$ sq. units. Using integration, find the value of m .

TYPE 3e Area of Circular/Elliptical Regions – Sector Area Parameter Determination

Given: Line $y = mx$ ($m > 0$), circle $x^2 + y^2 = 4$, x -axis in 1st quadrant. Given Area $= \frac{\pi}{2}$.

To Find: Value of parameter m .



Sector of circle $x^2 + y^2 = 4$ of area $\pi/2$

Concept / Formula Used: Area of Circular Sector $= \frac{1}{2}R^2\theta = 2\theta$

Solution:

Intersection point x_0 : $x^2 + (mx)^2 = 4 \implies x_0 = \frac{2}{\sqrt{1+m^2}}$. Using polar coordinates/angle method: the line $y = mx$ makes an angle $\theta = \tan^{-1}(m)$ with the x -axis. The area of a circular sector of radius $R = 2$ and central angle θ is:

$$\begin{aligned} \text{Area} &= \frac{1}{2}R^2\theta \\ &= \frac{1}{2}(4)\theta = 2\theta \end{aligned}$$

Given that Area $= \frac{\pi}{2}$:

$$\begin{aligned} 2\theta &= \frac{\pi}{2} \\ \theta &= \frac{\pi}{4} \end{aligned}$$

Since $\theta = \tan^{-1}(m)$:

$$m = \tan\left(\frac{\pi}{4}\right) = 1$$

Final Answer

$$m = 1$$

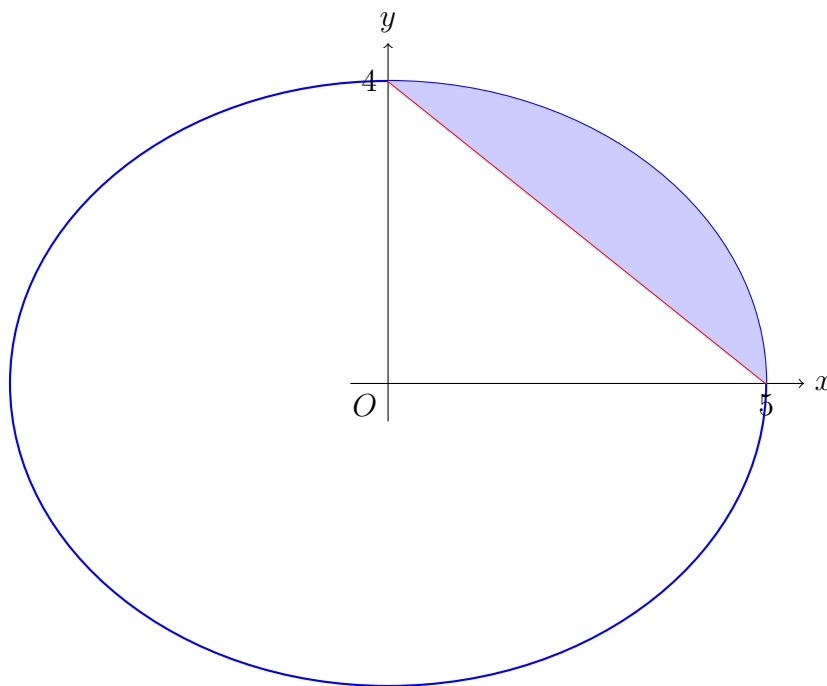
Question 21

Find the area of the smaller region bounded by the curves $\frac{x^2}{25} + \frac{y^2}{16} = 1$ and $\frac{x}{5} + \frac{y}{4} = 1$, using integration.

TYPE 21 Area Between Two Curves (Intersection-Bounded) – Ellipse Line Enclosed Area

Given: Ellipse $\frac{x^2}{25} + \frac{y^2}{16} = 1$, line $\frac{x}{5} + \frac{y}{4} = 1$.

To Find: Area of the smaller region between ellipse and line.



Smaller region between ellipse and straight line

Concept / Formula Used: Area = $\int_0^5 (y_{\text{ellipse}} - y_{\text{line}}) dx$

Solution:

Express y for ellipse and line:

$$y_{\text{ellipse}} = \frac{4}{5} \sqrt{25 - x^2}$$

$$y_{\text{line}} = 4 \left(1 - \frac{x}{5} \right) = \frac{4}{5} (5 - x)$$

The area of the smaller region is:

$$\begin{aligned}
 A &= \int_0^5 \frac{4}{5} \left(\sqrt{25 - x^2} - (5 - x) \right) dx \\
 &= \frac{4}{5} \left[\frac{x}{2} \sqrt{25 - x^2} + \frac{25}{2} \sin^{-1} \left(\frac{x}{5} \right) - 5x + \frac{x^2}{2} \right]_0^5 \\
 &= \frac{4}{5} \left(\left(0 + \frac{25}{2} \left(\frac{\pi}{2} \right) - 25 + \frac{25}{2} \right) - 0 \right) \\
 &= \frac{4}{5} \left(\frac{25\pi}{4} - \frac{25}{2} \right) \\
 &= 5\pi - 10 \text{ sq. units}
 \end{aligned}$$

Final Answer

Area = $(5\pi - 10)$ sq. units

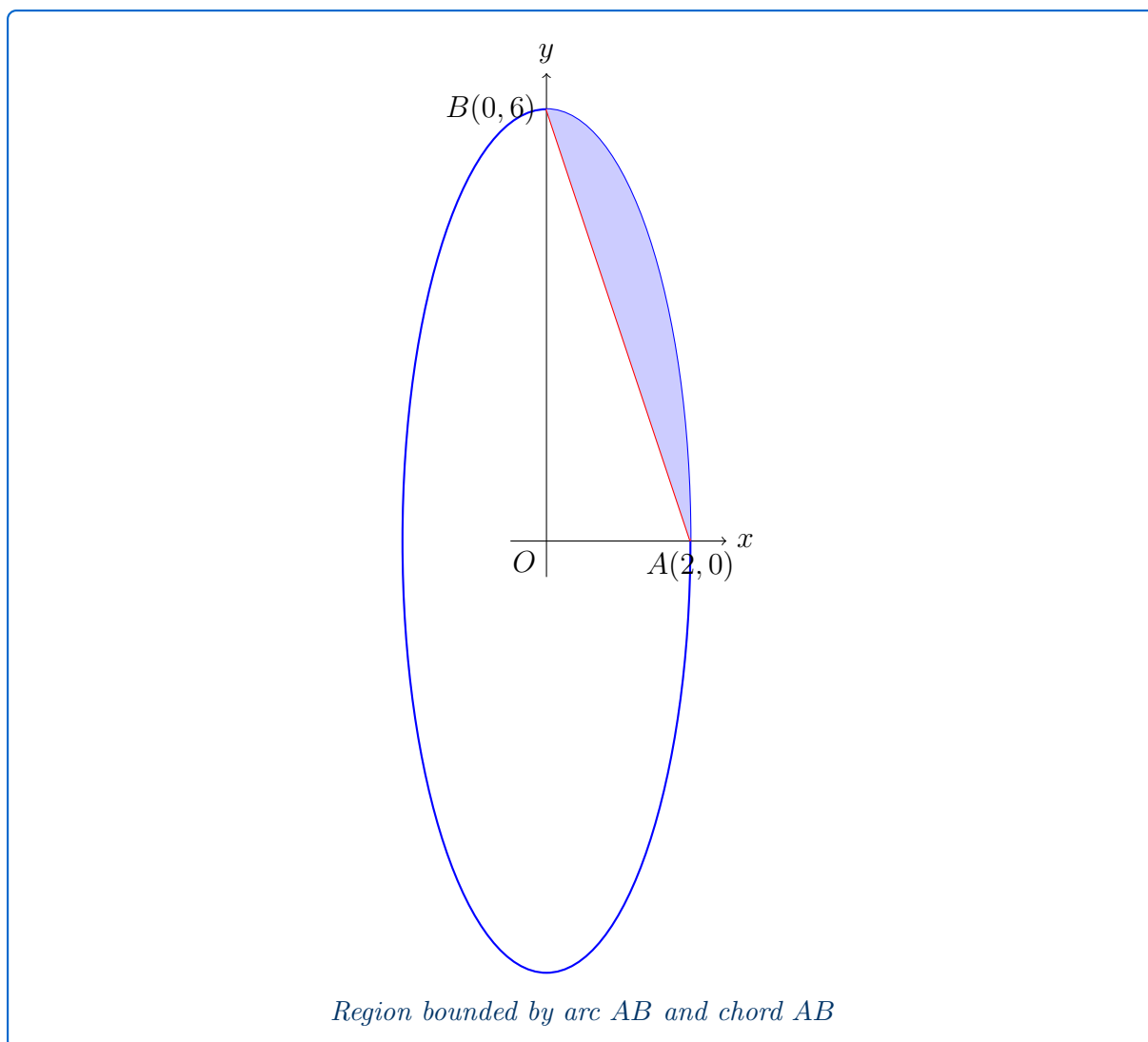
Question 22

AOB is a part of the ellipse $9x^2 + y^2 = 36$ in the first quadrant such that $OA = 2$ and $OB = 6$. Find the area enclosed by the arc AB and the chord AB.

TYPE 21 Area Between Two Curves (Intersection-Bounded) – Ellipse Line Enclosed Area

Given: Ellipse $9x^2 + y^2 = 36 \implies \frac{x^2}{4} + \frac{y^2}{36} = 1$, $OA = 2$, $OB = 6$.

To Find: Area enclosed by arc AB and chord AB .



Concept / Formula Used: Area = $\int_0^2 (y_{\text{ellipse}} - y_{\text{chord}}) dx$

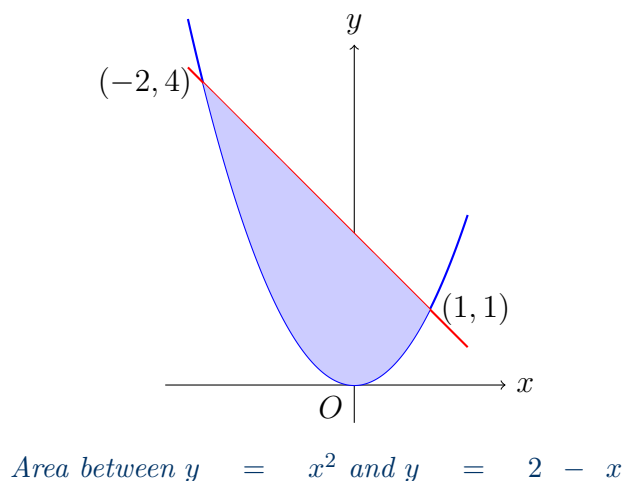
Solution:

From ellipse equation: $y_{\text{ellipse}} = \sqrt{36 - 9x^2} = 3\sqrt{4 - x^2}$. Equation of chord AB joining (2, 0) and (0, 6):

$$\frac{x}{2} + \frac{y}{6} = 1 \implies y = 6 - 3x$$

Area enclosed is:

$$\begin{aligned} A &= \int_0^2 (3\sqrt{4 - x^2} - (6 - 3x)) dx \\ &= 3 \int_0^2 \sqrt{4 - x^2} dx - \int_0^2 (6 - 3x) dx \\ &= 3 \left[\frac{x}{2} \sqrt{4 - x^2} + 2 \sin^{-1} \left(\frac{x}{2} \right) \right]_0^2 - \left[6x - \frac{3x^2}{2} \right]_0^2 \\ &= 3 \left(2 \cdot \frac{\pi}{2} \right) - (12 - 6) \\ &= 3\pi - 6 \text{ sq. units} \end{aligned}$$

Final AnswerArea = $(3\pi - 6)$ sq. units**Question 23 (i)**Draw a rough sketch and find the area bounded by the curve $x^2 = y$ and $x + y = 2$.**TYPE 2c Area Between Two Curves (Intersection-Bounded) – Parabola Line Enclosed Area****★ Likely Exam Question****Given:** Parabola $y = x^2$, line $y = 2 - x$.**To Find:** Rough sketch and bounded area.**Concept / Formula Used:** Area = $\int_{x_1}^{x_2} (y_{\text{line}} - y_{\text{parabola}}) dx$ **Solution:**

Find intersection points:

$$\begin{aligned}
 x^2 &= 2 - x \\
 x^2 + x - 2 &= 0 \\
 (x + 2)(x - 1) &= 0 \implies x = -2 \quad \text{or} \quad x = 1
 \end{aligned}$$

Evaluating the area integral:

$$\begin{aligned} A &= \int_{-2}^1 (2 - x - x^2) dx \\ &= \left[2x - \frac{x^2}{2} - \frac{x^3}{3} \right]_{-2}^1 \\ &= \left(2 - \frac{1}{2} - \frac{1}{3} \right) - \left(-4 - \frac{4}{2} - \frac{-8}{3} \right) \\ &= \frac{7}{6} - \left(-6 + \frac{8}{3} \right) \\ &= \frac{7}{6} - \left(-\frac{10}{3} \right) \\ &= \frac{7+20}{6} = \frac{27}{6} = \frac{9}{2} \text{ sq. units} \end{aligned}$$

Final Answer

$$\text{Area} = \frac{9}{2} \text{ sq. units}$$

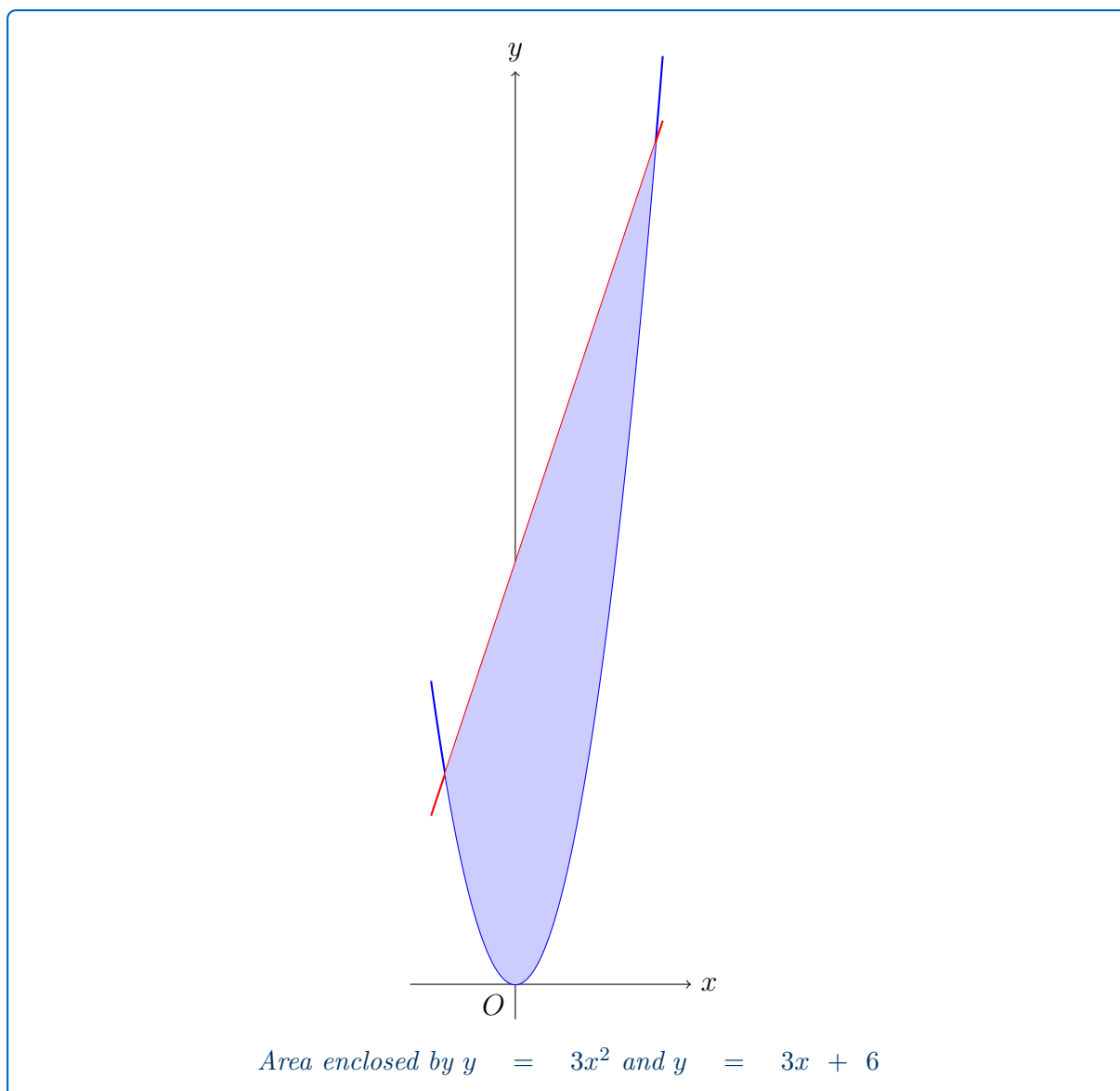
Question 23 (ii)

Using integration, find the area of the region enclosed by the parabola $y = 3x^2$ and the line $3x - y + 6 = 0$.

TYPE 2c Area Between Two Curves (Intersection-Bounded) – Parabola Line Enclosed Area

Given: Parabola $y = 3x^2$, line $y = 3x + 6$.

To Find: Enclosed area.



Concept / Formula Used: $\text{Area} = \int_{x_1}^{x_2} (y_{\text{line}} - y_{\text{parabola}}) dx$

Solution:

Find intersection points:

$$\begin{aligned} 3x^2 &= 3x + 6 \\ 3x^2 - 3x - 6 &= 0 \implies x^2 - x - 2 = 0 \\ (x - 2)(x + 1) &= 0 \implies x = -1, 2 \end{aligned}$$

Calculating integral:

$$\begin{aligned}
 A &= \int_{-1}^2 (3x + 6 - 3x^2) dx \\
 &= 3 \int_{-1}^2 (x + 2 - x^2) dx \\
 &= 3 \left[\frac{x^2}{2} + 2x - \frac{x^3}{3} \right]_{-1}^2 \\
 &= 3 \left[\left(2 + 4 - \frac{8}{3} \right) - \left(\frac{1}{2} - 2 + \frac{1}{3} \right) \right] \\
 &= 3 \left[\frac{10}{3} - \left(-\frac{7}{6} \right) \right] \\
 &= 3 \left[\frac{20 + 7}{6} \right] = \frac{27}{2} \text{ sq. units}
 \end{aligned}$$

Final Answer

$$\text{Area} = \frac{27}{2} \text{ sq. units}$$

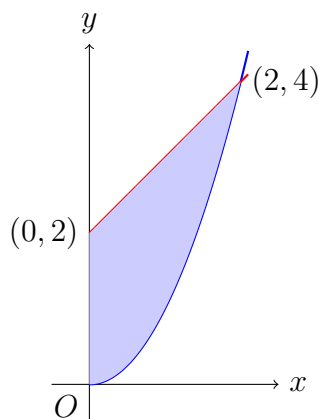
Question 23 (iii)

Using integration, find the area of the region bounded by the line $x - y + 2 = 0$, the curve $x = \sqrt{y}$ and y -axis.

TYPE 2c Area Between Two Curves (Intersection-Bounded) – Parabola Line Enclosed Area

Given: Line $y = x + 2$, curve $x = \sqrt{y} \implies y = x^2$ ($x \geq 0$), y -axis ($x = 0$).

To Find: Area of the region.



Region between line $y = x + 2$, $y = x^2$, and y -axis

Concept / Formula Used: Area = $\int_0^2 (y_{\text{line}} - y_{\text{curve}}) dx$

Solution:

Intersection point for $x \geq 0$:

$$\begin{aligned} x^2 &= x + 2 \\ x^2 - x - 2 &= 0 \implies (x - 2)(x + 1) = 0 \implies x = 2 \quad (x \geq 0) \end{aligned}$$

Integrating from $x = 0$ to $x = 2$:

$$\begin{aligned} A &= \int_0^2 ((x + 2) - x^2) dx \\ &= \left[\frac{x^2}{2} + 2x - \frac{x^3}{3} \right]_0^2 \\ &= \left(\frac{4}{2} + 4 - \frac{8}{3} \right) - 0 \\ &= 6 - \frac{8}{3} \\ &= \frac{10}{3} \text{ sq. units} \end{aligned}$$

Final Answer

$$\text{Area} = \frac{10}{3} \text{ sq. units}$$

Question 24 (i)

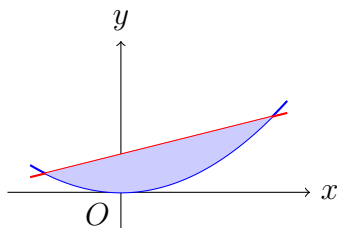
Find the area bounded by the curve $x^2 = 4y$ and the straight line $x = 4y - 2$.

TYPE 2c Area Between Two Curves (Intersection-Bounded) – Parabola Line Enclosed Area

★ Likely Exam Question

Given: Parabola $x^2 = 4y$, line $x = 4y - 2 \implies 4y = x + 2$.

To Find: Area enclosed.



Area bounded by $x^2 = 4y$ and $x = 4y - 2$

Concept / Formula Used: Area = $\int_{x_1}^{x_2} (y_{\text{line}} - y_{\text{parabola}}) dx$

Solution:

Equate $4y$ from both equations:

$$x^2 = x + 2$$

$$x^2 - x - 2 = 0 \implies (x - 2)(x + 1) = 0 \implies x = -1, 2$$

Calculate area:

$$\begin{aligned}
 A &= \int_{-1}^2 \left(\frac{x+2}{4} - \frac{x^2}{4} \right) dx \\
 &= \frac{1}{4} \int_{-1}^2 (x + 2 - x^2) dx \\
 &= \frac{1}{4} \left[\frac{x^2}{2} + 2x - \frac{x^3}{3} \right]_{-1}^2 \\
 &= \frac{1}{4} \left(\frac{10}{3} - \left(-\frac{7}{6} \right) \right) \\
 &= \frac{1}{4} \left(\frac{27}{6} \right) = \frac{9}{8} \text{ sq. units}
 \end{aligned}$$

Final Answer

$$\text{Area} = \frac{9}{8} \text{ sq. units}$$

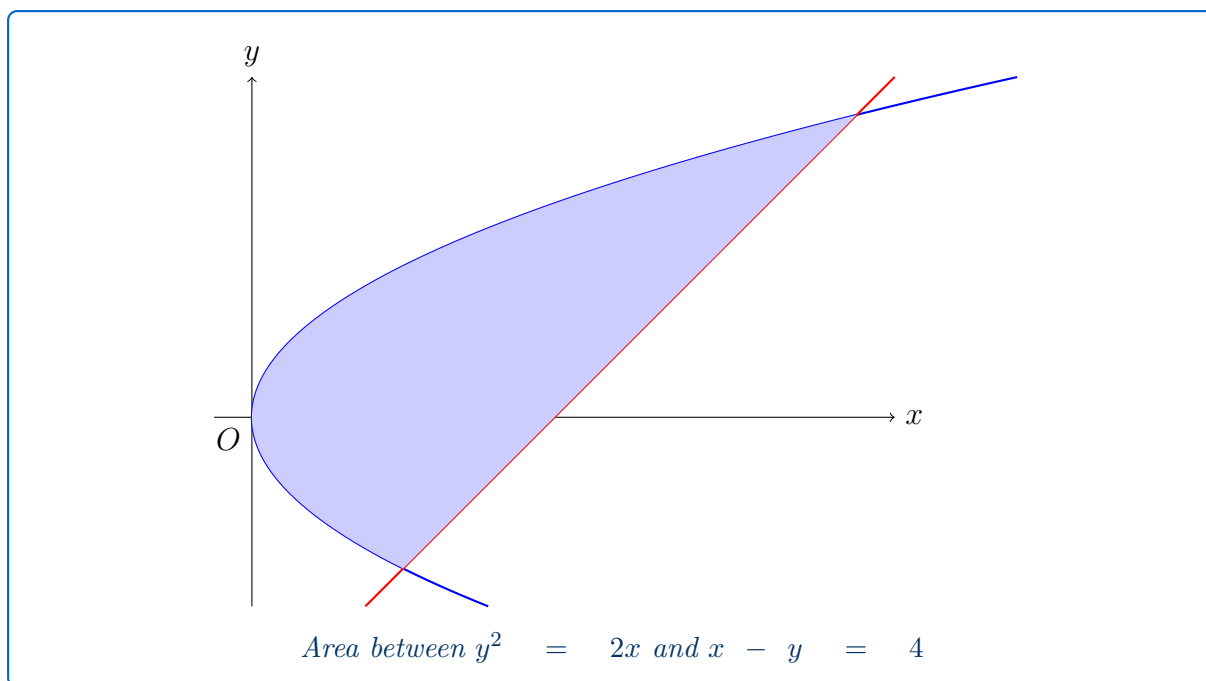
Question 24 (ii)

Find the area of the region bounded by the parabola $y^2 = 2x$ and the straight line $x - y = 4$.

TYPE 1b Area Under a Single Curve – Horizontal Strip Integration

Given: Parabola $y^2 = 2x \implies x = \frac{y^2}{2}$, line $x - y = 4 \implies x = y + 4$.

To Find: Enclosed area.



Concept / Formula Used: Area = $\int_{y_1}^{y_2} (x_{\text{line}} - x_{\text{parabola}}) dy$

Solution:

Find intersection y -values:

$$\begin{aligned} \frac{y^2}{2} &= y + 4 \\ y^2 - 2y - 8 &= 0 \implies (y - 4)(y + 2) = 0 \implies y = -2, 4 \end{aligned}$$

Integrating with respect to y :

$$\begin{aligned} A &= \int_{-2}^4 \left((y + 4) - \frac{y^2}{2} \right) dy \\ &= \left[\frac{y^2}{2} + 4y - \frac{y^3}{6} \right]_{-2}^4 \\ &= \left(\frac{16}{2} + 16 - \frac{64}{6} \right) - \left(\frac{4}{2} - 8 - \frac{-8}{6} \right) \\ &= \left(24 - \frac{32}{3} \right) - \left(-6 + \frac{4}{3} \right) \\ &= \frac{40}{3} - \left(-\frac{14}{3} \right) \\ &= \frac{54}{3} = 18 \text{ sq. units} \end{aligned}$$

Final Answer

Area = 18 sq. units

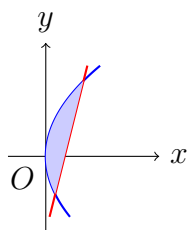
Question 24 (iii)

Find the area of the region $\{(x, y) : y^2 \leq 2x, y \geq 4x - 1\}$, by using integration.

TYPE 1b Area Under a Single Curve – Horizontal Strip Integration

Given: Region bounded by $y^2 = 2x \implies x = \frac{y^2}{2}$ and line $y = 4x - 1 \implies x = \frac{y+1}{4}$.

To Find: Enclosed area.



Region defined by inequalities $y^2 \leq 2x$ and $y \geq 4x - 1$

Concept / Formula Used: Area $= \int_{y_1}^{y_2} (x_{\text{line}} - x_{\text{parabola}}) dy$

Solution:

Find intersection values for y :

$$\begin{aligned}\frac{y^2}{2} &= \frac{y+1}{4} \\ 2y^2 &= y+1 \implies 2y^2 - y - 1 = 0 \\ (2y+1)(y-1) &= 0 \implies y = -\frac{1}{2}, 1\end{aligned}$$

Integrate with respect to y :

$$\begin{aligned}A &= \int_{-1/2}^1 \left(\frac{y+1}{4} - \frac{y^2}{2} \right) dy \\ &= \frac{1}{4} \left[\frac{y^2}{2} + y - \frac{2y^3}{3} \right]_{-1/2}^1 \\ &= \frac{1}{4} \left[\left(\frac{1}{2} + 1 - \frac{2}{3} \right) - \left(\frac{1}{8} - \frac{1}{2} + \frac{1}{12} \right) \right] \\ &= \frac{1}{4} \left[\frac{5}{6} - \left(-\frac{7}{24} \right) \right] \\ &= \frac{1}{4} \left[\frac{20+7}{24} \right] = \frac{27}{96} = \frac{9}{32} \text{ sq. units}\end{aligned}$$

Final Answer

$$\text{Area} = \frac{9}{32} \text{ sq. units}$$

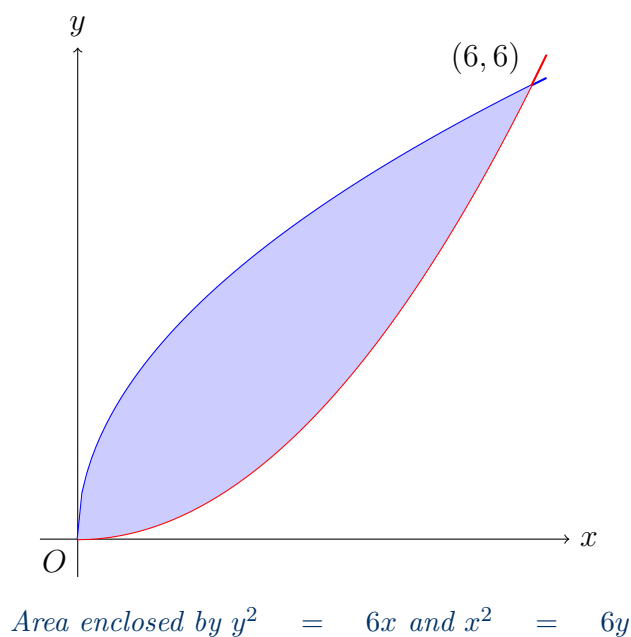
Question 25 (i)

Draw a rough sketch and find the area of the region bounded by the two parabolas $y^2 = 6x$ and $x^2 = 6y$.

TYPE 2i Area Between Two Curves (Intersection-Bounded) – Two Parabolas Enclosed Area

Given: Parabolas $y^2 = 6x$ and $x^2 = 6y$.

To Find: Sketch and bounded area.



Concept / Formula Used: Area = $\int_0^a \left(\sqrt{ax} - \frac{x^2}{a} \right) dx = \frac{a^2}{3}$

Solution:

Find intersection points:

$$y = \frac{x^2}{6} \implies \left(\frac{x^2}{6} \right)^2 = 6x$$

$$x^4 = 216x \implies x(x^3 - 216) = 0 \implies x = 0, 6$$

Evaluating the area:

$$\begin{aligned}
 A &= \int_0^6 \left(\sqrt{6x} - \frac{x^2}{6} \right) dx \\
 &= \left[\sqrt{6} \cdot \frac{2}{3} x^{3/2} - \frac{x^3}{18} \right]_0^6 \\
 &= \left(\sqrt{6} \cdot \frac{2}{3} (6\sqrt{6}) - \frac{216}{18} \right) - 0 \\
 &= (24 - 12) \\
 &= 12 \text{ sq. units}
 \end{aligned}$$

Final Answer

Area = 12 sq. units

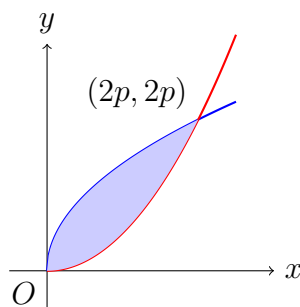
Question 25 (ii)

Find the area of the region bounded by the parabolas $y^2 = 2px$ and $x^2 = 2py$.

TYPE 2i Area Between Two Curves (Intersection-Bounded) – Two Parabolas Enclosed Area

Given: Parabolas $y^2 = 2px$ and $x^2 = 2py$.

To Find: Area enclosed between the two parabolas.



Area enclosed by $y^2 = 2px$ and $x^2 = 2py$

Concept / Formula Used: Area = $\int_0^{2p} \left(\sqrt{2px} - \frac{x^2}{2p} \right) dx$

Solution:

Find intersection points:

$$\begin{aligned}
 y = \frac{x^2}{2p} &\implies \left(\frac{x^2}{2p} \right)^2 = 2px \\
 \frac{x^4}{4p^2} &= 2px \implies x^4 = 8p^3x \\
 x(x^3 - 8p^3) &= 0 \implies x = 0, 2p
 \end{aligned}$$

Integrating:

$$\begin{aligned}
 A &= \int_0^{2p} \left(\sqrt{2p}\sqrt{x} - \frac{x^2}{2p} \right) dx \\
 &= \left[\sqrt{2p} \cdot \frac{2}{3} x^{3/2} - \frac{x^3}{6p} \right]_0^{2p} \\
 &= \sqrt{2p} \cdot \frac{2}{3} (2p)^{3/2} - \frac{(2p)^3}{6p} \\
 &= \frac{2}{3} (2p)^2 - \frac{8p^3}{6p} \\
 &= \frac{8p^2}{3} - \frac{4p^2}{3} \\
 &= \frac{4p^2}{3} \text{ sq. units}
 \end{aligned}$$

Final Answer

$$\text{Area} = \frac{4p^2}{3} \text{ sq. units}$$

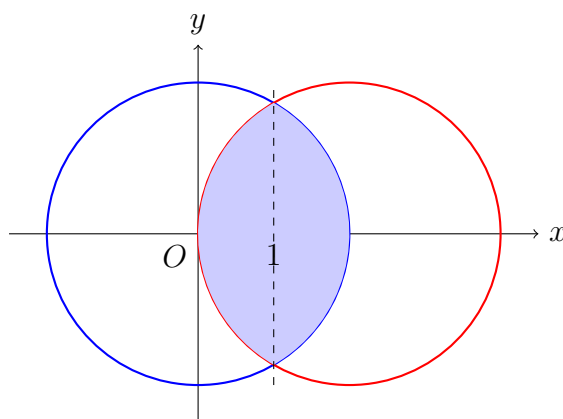
Question 26 (i)

Using integration, find the area of the region enclosed between the circles $x^2 + y^2 = 4$ and $(x - 2)^2 + y^2 = 4$.

TYPE 3f Area of Circular/Elliptical Regions – Intersecting Circles Area

Given: Circles $C_1 : x^2 + y^2 = 4$ and $C_2 : (x - 2)^2 + y^2 = 4$.

To Find: Area enclosed between the two circles.



Area enclosed by two overlapping circles of radius 2

Concept / Formula Used: Area = $2 \left[\int_0^1 \sqrt{4 - (x - 2)^2} dx + \int_1^2 \sqrt{4 - x^2} dx \right]$

Solution:

Find intersection x -coordinate:

$$\begin{aligned}x^2 + y^2 &= (x - 2)^2 + y^2 \\x^2 &= x^2 - 4x + 4 \implies 4x = 4 \implies x = 1\end{aligned}$$

By symmetry, total area $A = 4 \times$ area in first quadrant for C_1 from $x = 1$ to 2:

$$\begin{aligned}A &= 4 \int_1^2 \sqrt{4 - x^2} dx \\&= 4 \left[\frac{x}{2} \sqrt{4 - x^2} + 2 \sin^{-1} \left(\frac{x}{2} \right) \right]_1^2 \\&= 4 \left[(0 + 2 \sin^{-1}(1)) - \left(\frac{\sqrt{3}}{2} + 2 \sin^{-1} \left(\frac{1}{2} \right) \right) \right] \\&= 4 \left[\pi - \left(\frac{\sqrt{3}}{2} + \frac{\pi}{3} \right) \right] \\&= 4 \left[\frac{2\pi}{3} - \frac{\sqrt{3}}{2} \right] \\&= \frac{8\pi}{3} - 2\sqrt{3} \text{ sq. units}\end{aligned}$$

Final Answer

$$\text{Area} = \left(\frac{8\pi}{3} - 2\sqrt{3} \right) \text{ sq. units}$$

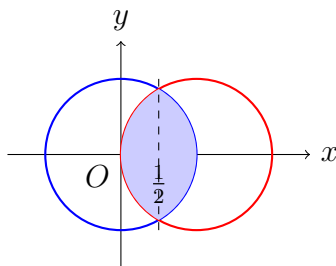
Question 26 (ii)

Using integration, find the area of the region enclosed between the circles $x^2 + y^2 = 1$ and $(x - 1)^2 + y^2 = 1$.

TYPE 3f Area of Circular/Elliptical Regions – Intersecting Circles Area

Given: Circles $C_1 : x^2 + y^2 = 1$ and $C_2 : (x - 1)^2 + y^2 = 1$.

To Find: Area enclosed between the two circles.



Area enclosed by two overlapping unit circles

Concept / Formula Used: Area = $4 \int_{1/2}^1 \sqrt{1-x^2} dx$

Solution:

Find intersection x -coordinate:

$$\begin{aligned} x^2 + y^2 &= (x-1)^2 + y^2 \\ x^2 &= x^2 - 2x + 1 \implies 2x = 1 \implies x = \frac{1}{2} \end{aligned}$$

By symmetry, area $A = 4 \times \int_{1/2}^1 \sqrt{1-x^2} dx$:

$$\begin{aligned} A &= 4 \left[\frac{x}{2} \sqrt{1-x^2} + \frac{1}{2} \sin^{-1}(x) \right]_{1/2}^1 \\ &= 4 \left[\left(0 + \frac{1}{2} \sin^{-1}(1) \right) - \left(\frac{1}{4} \sqrt{1-\frac{1}{4}} + \frac{1}{2} \sin^{-1} \left(\frac{1}{2} \right) \right) \right] \\ &= 4 \left[\frac{\pi}{4} - \left(\frac{\sqrt{3}}{8} + \frac{\pi}{12} \right) \right] \\ &= 4 \left[\frac{\pi}{6} - \frac{\sqrt{3}}{8} \right] \\ &= \frac{2\pi}{3} - \frac{\sqrt{3}}{2} \text{ sq. units} \end{aligned}$$

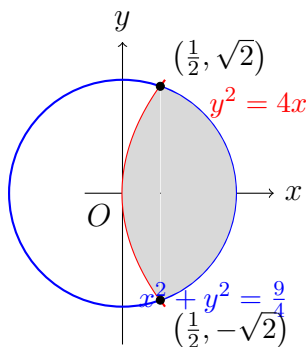
Final Answer

$$\text{Area} = \left(\frac{2\pi}{3} - \frac{\sqrt{3}}{2} \right) \text{ sq. units}$$

Question 27 (i)

Find the area of the region $\{(x, y) : y^2 \leq 4x, 4x^2 + 4y^2 \leq 9\}$ using the method of integration.

TYPE 2n Area Between Two Curves (Intersection-Bounded) – Area Between Parabola and Circle



Region bounded by parabola $y^2 = 4x$ and circle $x^2 + y^2 = \frac{9}{4}$

Given: Curves $y^2 = 4x$ and $4x^2 + 4y^2 = 9$

To Find: Area of the enclosed region

Concept / Formula Used: $\int \sqrt{a^2 - x^2} dx = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1}\left(\frac{x}{a}\right) + c$

Solution:

Rewrite the equation of the circle in standard form:

$$4x^2 + 4y^2 = 9$$

$$x^2 + y^2 = \frac{9}{4}$$

To find the points of intersection of $y^2 = 4x$ and $x^2 + y^2 = \frac{9}{4}$, substitute $y^2 = 4x$ into the circle's equation:

$$x^2 + 4x = \frac{9}{4}$$

$$4x^2 + 16x - 9 = 0$$

$$4x^2 + 18x - 2x - 9 = 0$$

$$2x(2x + 9) - 1(2x + 9) = 0$$

$$(2x - 1)(2x + 9) = 0$$

Since $x \geq 0$ for $y^2 = 4x$, we take:

$$x = \frac{1}{2}$$

Due to symmetry about the x -axis, the required area is twice the area in the first quadrant:

$$\begin{aligned} \text{Area} &= 2 \left[\int_0^{1/2} y_{\text{parabola}} dx + \int_{1/2}^{3/2} y_{\text{circle}} dx \right] \\ &= 2 \left[\int_0^{1/2} 2\sqrt{x} dx + \int_{1/2}^{3/2} \sqrt{\frac{9}{4} - x^2} dx \right] \end{aligned}$$

Evaluating the first integral:

$$\begin{aligned} \int_0^{1/2} 2\sqrt{x} dx &= 2 \left[\frac{2}{3} x^{3/2} \right]_0^{1/2} \\ &= \frac{4}{3} \left(\frac{1}{2} \right)^{3/2} \\ &= \frac{4}{3} \cdot \frac{1}{2\sqrt{2}} \\ &= \frac{\sqrt{2}}{3} \end{aligned}$$

Using the standard result $\int \sqrt{a^2 - x^2} dx = \frac{x}{2}\sqrt{a^2 - x^2} + \frac{a^2}{2}\sin^{-1}\left(\frac{x}{a}\right) + c$ for the second integral with $a = \frac{3}{2}$:

$$\begin{aligned}\int_{1/2}^{3/2} \sqrt{\left(\frac{3}{2}\right)^2 - x^2} dx &= \left[\frac{x}{2}\sqrt{\frac{9}{4} - x^2} + \frac{9}{8}\sin^{-1}\left(\frac{2x}{3}\right) \right]_{1/2}^{3/2} \\ &= \left(0 + \frac{9}{8}\sin^{-1}(1) \right) - \left(\frac{1}{4}\sqrt{\frac{9}{4} - \frac{1}{4}} + \frac{9}{8}\sin^{-1}\left(\frac{1}{3}\right) \right) \\ &= \frac{9}{8} \cdot \frac{\pi}{2} - \frac{1}{4}\sqrt{2} - \frac{9}{8}\sin^{-1}\left(\frac{1}{3}\right) \\ &= \frac{9\pi}{16} - \frac{\sqrt{2}}{4} - \frac{9}{8}\sin^{-1}\left(\frac{1}{3}\right)\end{aligned}$$

Adding both integrals together and multiplying by 2:

$$\begin{aligned}\text{Area} &= 2 \left[\frac{\sqrt{2}}{3} + \frac{9\pi}{16} - \frac{\sqrt{2}}{4} - \frac{9}{8}\sin^{-1}\left(\frac{1}{3}\right) \right] \\ &= 2 \left[\frac{\sqrt{2}}{12} + \frac{9\pi}{16} - \frac{9}{8}\sin^{-1}\left(\frac{1}{3}\right) \right] \\ &= \frac{\sqrt{2}}{6} + \frac{9\pi}{8} - \frac{9}{4}\sin^{-1}\left(\frac{1}{3}\right)\end{aligned}$$

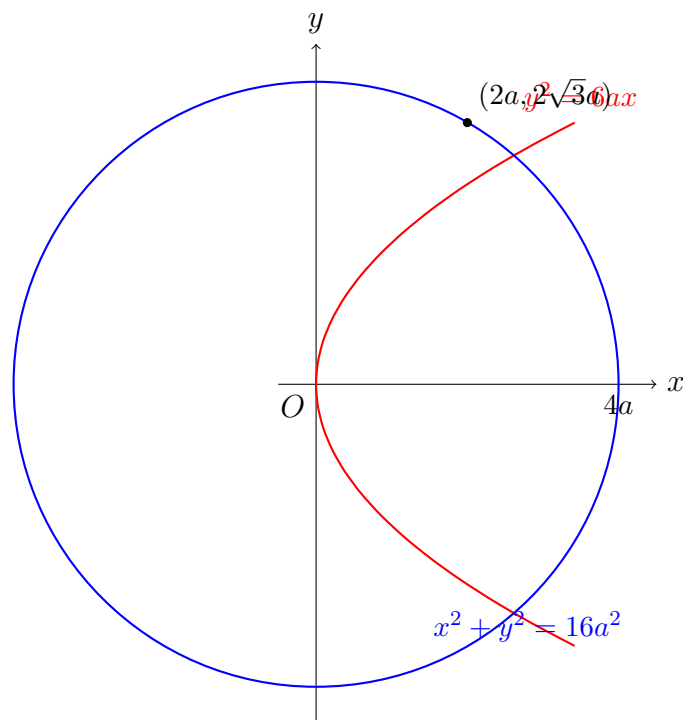
Final Answer

$$\text{Area} = \left(\frac{\sqrt{2}}{6} + \frac{9\pi}{8} - \frac{9}{4}\sin^{-1}\left(\frac{1}{3}\right) \right) \text{ sq. units}$$

Question 27 (ii)

Using integration, find the area of the region $\{(x, y) : y^2 \leq 6ax, x^2 + y^2 \leq 16a^2\}$.

TYPE 2n Area Between Two Curves (Intersection-Bounded) – Area Between Parabola and Circle



Region bounded by $y^2 = 6ax$ and $x^2 + y^2 = 16a^2$

Given: Parabola $y^2 = 6ax$ and circle $x^2 + y^2 = 16a^2$

To Find: Area of the region

Concept / Formula Used: $\int \sqrt{a^2 - x^2} dx = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \left(\frac{x}{a} \right) + c$

Solution:

To find the intersection of $y^2 = 6ax$ and $x^2 + y^2 = 16a^2$, substitute $y^2 = 6ax$ into the circle equation:

$$\begin{aligned} x^2 + 6ax - 16a^2 &= 0 \\ (x + 8a)(x - 2a) &= 0 \end{aligned}$$

Since $x \geq 0$, we get:

$$x = 2a$$

By symmetry about the x -axis:

$$\text{Area} = 2 \left[\int_0^{2a} \sqrt{6ax} dx + \int_{2a}^{4a} \sqrt{16a^2 - x^2} dx \right]$$

Evaluating the first integral:

$$\begin{aligned}
 \int_0^{2a} \sqrt{6a} \cdot \sqrt{x} \, dx &= \sqrt{6a} \left[\frac{2}{3} x^{3/2} \right]_0^{2a} \\
 &= \sqrt{6a} \cdot \frac{2}{3} (2a)^{3/2} \\
 &= \sqrt{6a} \cdot \frac{2}{3} \cdot 2\sqrt{2a}\sqrt{a} \\
 &= \frac{8\sqrt{3}}{3} a^2
 \end{aligned}$$

Using the standard integral formula for the second integral with radius $4a$:

$$\begin{aligned}
 \int_{2a}^{4a} \sqrt{(4a)^2 - x^2} \, dx &= \left[\frac{x}{2} \sqrt{16a^2 - x^2} + \frac{16a^2}{2} \sin^{-1} \left(\frac{x}{4a} \right) \right]_{2a}^{4a} \\
 &= \left(0 + 8a^2 \cdot \frac{\pi}{2} \right) - \left(\frac{2a}{2} \sqrt{16a^2 - 4a^2} + 8a^2 \sin^{-1} \left(\frac{1}{2} \right) \right) \\
 &= 4\pi a^2 - a\sqrt{12a^2} - 8a^2 \cdot \frac{\pi}{6} \\
 &= 4\pi a^2 - 2\sqrt{3}a^2 - \frac{4\pi}{3}a^2 \\
 &= \frac{8\pi}{3}a^2 - 2\sqrt{3}a^2
 \end{aligned}$$

Total Area:

$$\begin{aligned}
 \text{Area} &= 2 \left[\frac{8\sqrt{3}}{3}a^2 + \frac{8\pi}{3}a^2 - 2\sqrt{3}a^2 \right] \\
 &= 2 \left[\frac{2\sqrt{3}}{3}a^2 + \frac{8\pi}{3}a^2 \right] \\
 &= \frac{4}{3} (\sqrt{3} + 4\pi) a^2
 \end{aligned}$$

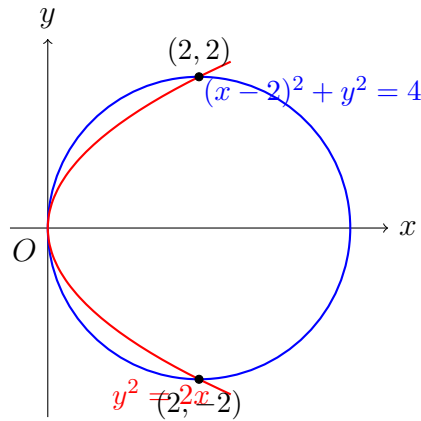
Final Answer

$$\text{Area} = \frac{4}{3} (\sqrt{3} + 4\pi) a^2 \text{ sq. units}$$

Question 28

Find the area of the smaller region bounded by the curves $y^2 = 2x$ and $x^2 + y^2 = 4x$.

TYPE 2n Area Between Two Curves (Intersection-Bounded) – Area Between Parabola and Circle



Bounded region between $y^2 = 2x$ and circle $x^2 + y^2 = 4x$

Given: Parabola $y^2 = 2x$ and circle $x^2 + y^2 = 4x \implies (x-2)^2 + y^2 = 4$

To Find: Area of the smaller region bounded by both curves

Concept / Formula Used: $\int \sqrt{a^2 - u^2} du = \frac{u}{2} \sqrt{a^2 - u^2} + \frac{a^2}{2} \sin^{-1} \left(\frac{u}{a} \right) + c$

Solution:

Find the points of intersection by solving $y^2 = 2x$ and $x^2 + y^2 = 4x$:

$$x^2 + 2x = 4x$$

$$x^2 - 2x = 0$$

$$x(x-2) = 0$$

Thus, $x = 0$ or $x = 2$. By symmetry about the x -axis, the required area is:

$$\begin{aligned} \text{Area} &= 2 \int_0^2 (y_{\text{circle}} - y_{\text{parabola}}) dx \\ &= 2 \int_0^2 \left(\sqrt{4x - x^2} - \sqrt{2x} \right) dx \end{aligned}$$

Completing the square inside the radical for the circle:

$$4x - x^2 = 4 - (x^2 - 4x + 4) = 2^2 - (x-2)^2$$

Substitute $u = x - 2$:

$$du = dx$$

When $x = 0$, $u = -2$; when $x = 2$, $u = 0$.

$$\begin{aligned}\int_0^2 \sqrt{4x - x^2} dx &= \int_{-2}^0 \sqrt{2^2 - u^2} du \\ &= \left[\frac{u}{2} \sqrt{4 - u^2} + \frac{4}{2} \sin^{-1} \left(\frac{u}{2} \right) \right]_{-2}^0 \\ &= (0 + 2 \sin^{-1}(0)) - (0 + 2 \sin^{-1}(-1)) \\ &= -2 \left(-\frac{\pi}{2} \right) \\ &= \pi\end{aligned}$$

Evaluating the parabola integral:

$$\begin{aligned}\int_0^2 \sqrt{2x} dx &= \sqrt{2} \left[\frac{2}{3} x^{3/2} \right]_0^2 \\ &= \sqrt{2} \cdot \frac{2}{3} \cdot 2\sqrt{2} \\ &= \frac{8}{3}\end{aligned}$$

Subtracting the two values:

$$\text{Area} = 2 \left(\pi - \frac{8}{3} \right)$$

Final Answer

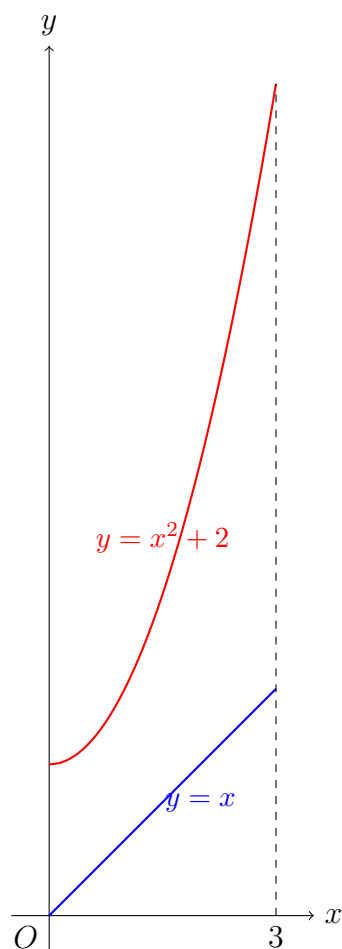
Area = $2 \left(\pi - \frac{8}{3} \right)$ sq. units

Question 29

Find the area of the region bounded by the parabola $y = x^2 + 2$ and the lines $y = x$, $x = 0$ and $x = 3$.

TYPE 2b Area Between Two Curves (Intersection-Bounded) – Area Between Two Curves

★ Likely Exam Question



Region between $y = x^2 + 2$ and $y = x$ from $x = 0$ to $x = 3$

Given: Curves $y = x^2 + 2$, $y = x$, and vertical lines $x = 0$, $x = 3$

To Find: Area of the bounded region

Concept / Formula Used: $\text{Area} = \int_a^b (y_{\text{upper}} - y_{\text{lower}}) dx$

Solution:

For all $x \in [0, 3]$, $x^2 + 2 > x$, so $y = x^2 + 2$ is the upper curve and $y = x$ is the lower

curve.

$$\begin{aligned}
 \text{Area} &= \int_0^3 ((x^2 + 2) - x) dx \\
 &= \int_0^3 (x^2 - x + 2) dx \\
 &= \left[\frac{x^3}{3} - \frac{x^2}{2} + 2x \right]_0^3 \\
 &= \left(\frac{3^3}{3} - \frac{3^2}{2} + 2(3) \right) - (0) \\
 &= 9 - \frac{9}{2} + 6 \\
 &= 15 - \frac{9}{2} \\
 &= \frac{21}{2} = 10\frac{1}{2}
 \end{aligned}$$

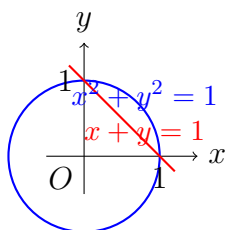
Final Answer

Area = $10\frac{1}{2}$ sq. units

Question 30

Find the area of the region $\{(x, y) : x^2 + y^2 \leq 1 \leq x + y\}$, using integration.

TYPE 2o Area Between Two Curves (Intersection-Bounded) – Area Bounded by Circle and Line



Region inside circle $x^2 + y^2 = 1$ and above line $x + y = 1$

Given: Region defined by $x^2 + y^2 \leq 1$ and $x + y \geq 1$

To Find: Area of the region

Concept / Formula Used: Area = $\int_0^1 (\sqrt{1-x^2} - (1-x)) dx$

Solution:

The intersection points of $x^2 + y^2 = 1$ and $x + y = 1$ are $(1, 0)$ and $(0, 1)$. The region is

in the first quadrant, bounded above by $y = \sqrt{1-x^2}$ and below by $y = 1-x$:

$$\begin{aligned}\text{Area} &= \int_0^1 \left(\sqrt{1-x^2} - (1-x) \right) dx \\ &= \int_0^1 \sqrt{1-x^2} dx - \int_0^1 (1-x) dx\end{aligned}$$

Using the standard formula $\int \sqrt{a^2 - x^2} dx = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1}\left(\frac{x}{a}\right) + c$:

$$\begin{aligned}\int_0^1 \sqrt{1-x^2} dx &= \left[\frac{x}{2} \sqrt{1-x^2} + \frac{1}{2} \sin^{-1}(x) \right]_0^1 \\ &= \left(0 + \frac{1}{2} \sin^{-1}(1) \right) - (0) \\ &= \frac{1}{2} \cdot \frac{\pi}{2} = \frac{\pi}{4}\end{aligned}$$

Evaluating the second integral:

$$\begin{aligned}\int_0^1 (1-x) dx &= \left[x - \frac{x^2}{2} \right]_0^1 \\ &= 1 - \frac{1}{2} = \frac{1}{2}\end{aligned}$$

Subtracting the results:

$$\text{Area} = \frac{\pi}{4} - \frac{1}{2} = \frac{1}{4}(\pi - 2)$$

Final Answer

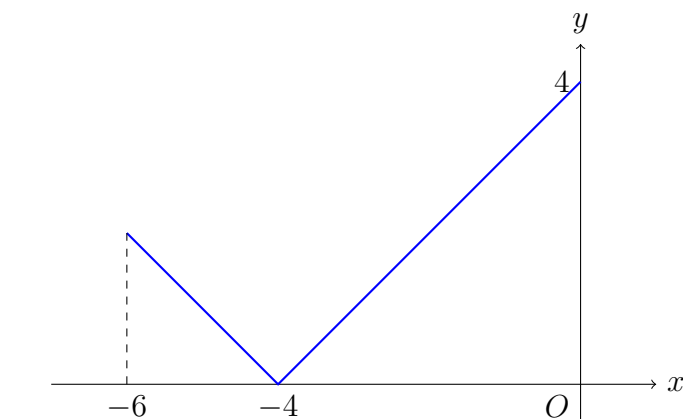
Area = $\frac{1}{4}(\pi - 2)$ sq. units

Question 31 (i)

Sketch the graph of $y = |x + 4|$. Using integration, find the area of the region under the curve $y = |x + 4|$ above x -axis and between $x = -6$ and $x = 0$.

TYPE 1g Area Under a Single Curve – Area Under Absolute Value Function

★ Likely Exam Question



Graph of $y = |x + 4|$ from $x = -6$ to $x = 0$

Given: Function $y = |x + 4|$ on interval $[-6, 0]$

To Find: Area under the curve above the x -axis

Concept / Formula Used: $|x + 4| = \begin{cases} -(x + 4), & x < -4 \\ x + 4, & x \geq -4 \end{cases}$

Solution:

By definition of absolute value:

$$y = \begin{cases} -(x + 4), & -6 \leq x < -4 \\ x + 4, & -4 \leq x \leq 0 \end{cases}$$

Split the integral at $x = -4$:

$$\begin{aligned} \text{Area} &= \int_{-6}^{-4} -(x + 4) dx + \int_{-4}^0 (x + 4) dx \\ &= \left[-\frac{x^2}{2} - 4x \right]_{-6}^{-4} + \left[\frac{x^2}{2} + 4x \right]_{-4}^0 \end{aligned}$$

Evaluating the first part:

$$\begin{aligned} \left[-\frac{x^2}{2} - 4x \right]_{-6}^{-4} &= \left(-\frac{(-4)^2}{2} - 4(-4) \right) - \left(-\frac{(-6)^2}{2} - 4(-6) \right) \\ &= (-8 + 16) - (-18 + 24) \\ &= 8 - 6 = 2 \end{aligned}$$

Evaluating the second part:

$$\begin{aligned} \left[\frac{x^2}{2} + 4x \right]_{-4}^0 &= (0) - \left(\frac{(-4)^2}{2} + 4(-4) \right) \\ &= -(8 - 16) = 8 \end{aligned}$$

Total Area:

$$\text{Area} = 2 + 8 = 10$$

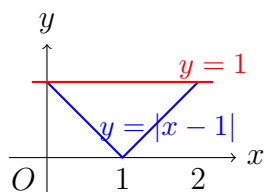
Final Answer

Area = 10 sq. units

Question 31 (ii)

Find the area bounded by the curves $y = |x - 1|$ and $y = 1$, using integration.

TYPE 2f Area Between Two Curves (Intersection-Bounded) – Area Between Line and Absolute Value



Region between $y = |x - 1|$ and $y = 1$

Given: Curves $y = |x - 1|$ and $y = 1$

To Find: Bounded area

Concept / Formula Used: $\text{Area} = \int_{x_1}^{x_2} (y_{\text{upper}} - y_{\text{lower}}) dx$

Solution:

Find the intersection of $y = |x - 1|$ and $y = 1$:

$$|x - 1| = 1$$

$$x - 1 = 1 \implies x = 2$$

$$x - 1 = -1 \implies x = 0$$

The upper boundary is $y = 1$ and the lower boundary is $y = |x - 1|$ on $[0, 2]$:

$$\begin{aligned} \text{Area} &= \int_0^2 (1 - |x - 1|) dx \\ &= \int_0^1 (1 - (1 - x)) dx + \int_1^2 (1 - (x - 1)) dx \\ &= \int_0^1 x dx + \int_1^2 (2 - x) dx \end{aligned}$$

Evaluating both integrals:

$$\int_0^1 x \, dx = \left[\frac{x^2}{2} \right]_0^1 = \frac{1}{2}$$

$$\int_1^2 (2-x) \, dx = \left[2x - \frac{x^2}{2} \right]_1^2 = (4-2) - \left(2 - \frac{1}{2} \right) = 2 - \frac{3}{2} = \frac{1}{2}$$

Total Area:

$$\text{Area} = \frac{1}{2} + \frac{1}{2} = 1$$

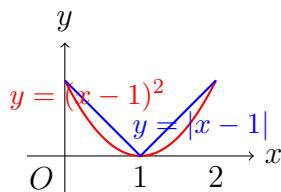
Final Answer

Area = 1 sq. unit

Question 32

Draw a rough sketch of the curves $y = (x-1)^2$ and $y = |x-1|$. Also find the area of the region bounded by these curves.

TYPE 2e Area Between Two Curves (Intersection-Bounded) – Area Between Parabola and Absolute Value



Region between parabola $y = (x-1)^2$ and $y = |x-1|$

Given: Curves $y = (x-1)^2$ and $y = |x-1|$

To Find: Area of the enclosed region

Concept / Formula Used: Area = $\int_0^2 (|x-1| - (x-1)^2) \, dx$

Solution:

To find the intersection points, set $(x-1)^2 = |x-1|$: Let $u = x-1$, then $u^2 = |u|$, giving $u = 0, 1, -1 \implies x = 1, 2, 0$. The region is symmetric about $x = 1$. Evaluating for $x \in [1, 2]$ and doubling:

$$\text{Area} = 2 \int_1^2 ((x-1) - (x-1)^2) \, dx$$

Substitute $u = x-1$:

$$du = dx$$

When $x = 1$, $u = 0$; when $x = 2$, $u = 1$.

$$\begin{aligned}\text{Area} &= 2 \int_0^1 (u - u^2) du \\ &= 2 \left[\frac{u^2}{2} - \frac{u^3}{3} \right]_0^1 \\ &= 2 \left(\frac{1}{2} - \frac{1}{3} \right) \\ &= 2 \left(\frac{1}{6} \right) = \frac{1}{3}\end{aligned}$$

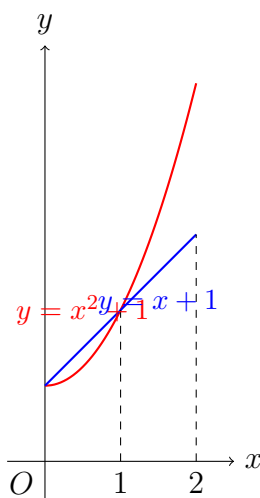
Final Answer

Area = $\frac{1}{3}$ sq. units

Question 33 (i)

Find the area of the region $\{(x, y) : 0 \leq y \leq x^2 + 1, 0 \leq y \leq x + 1, 0 \leq x \leq 2\}$.

TYPE 5a Area of Piecewise-Defined or Sign-Changing Regions – Piecewise Upper Boundary Area



Region under minimum of $y = x^2 + 1$ and $y = x + 1$ from $x = 0$ to $x = 2$

Given: Region bounded by $y \leq x^2 + 1$, $y \leq x + 1$, and $0 \leq x \leq 2$

To Find: Area of the specified region

Concept / Formula Used: Area = $\int_0^1 (x^2 + 1) dx + \int_1^2 (x + 1) dx$

Solution:

Find the intersection of $y = x^2 + 1$ and $y = x + 1$:

$$\begin{aligned}x^2 + 1 &= x + 1 \\x^2 - x &= 0 \implies x = 0 \text{ or } x = 1\end{aligned}$$

For $x \in [0, 1]$, $x^2 + 1 \leq x + 1$, so the upper boundary is $y = x^2 + 1$. For $x \in [1, 2]$, $x + 1 \leq x^2 + 1$, so the upper boundary is $y = x + 1$.

$$\begin{aligned}\text{Area} &= \int_0^1 (x^2 + 1) dx + \int_1^2 (x + 1) dx \\&= \left[\frac{x^3}{3} + x \right]_0^1 + \left[\frac{x^2}{2} + x \right]_1^2 \\&= \left(\frac{1}{3} + 1 \right) + \left(\left(\frac{4}{2} + 2 \right) - \left(\frac{1}{2} + 1 \right) \right) \\&= \frac{4}{3} + \left(4 - \frac{3}{2} \right) \\&= \frac{4}{3} + \frac{5}{2} \\&= \frac{8 + 15}{6} = \frac{23}{6}\end{aligned}$$

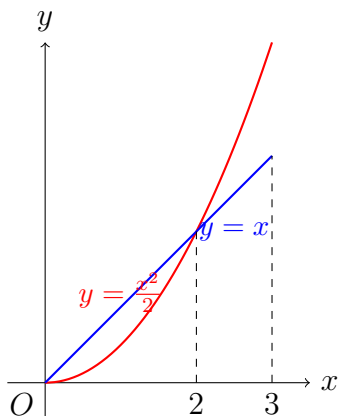
Final Answer

Area = $\frac{23}{6}$ sq. units

Question 33 (ii)

Using integration, find the area of the region $\{(x, y) : 0 \leq 2y \leq x^2, 0 \leq y \leq x, 0 \leq x \leq 3\}$.

TYPE 5a Area of Piecewise-Defined or Sign-Changing Regions – Piecewise Upper Boundary Area



Region $0 \leq y \leq \frac{x^2}{2}$ and $y \leq x$ for $0 \leq x \leq 3$

Given: Region bounded by $y \leq \frac{x^2}{2}$, $y \leq x$, and $0 \leq x \leq 3$

To Find: Area of the region

Concept / Formula Used: $\text{Area} = \int_0^2 \frac{x^2}{2} dx + \int_2^3 x dx$

Solution:

To determine which curve limits y , solve $\frac{x^2}{2} = x$:

$$x^2 - 2x = 0 \implies x = 0 \text{ or } x = 2$$

For $x \in [0, 2]$, $\frac{x^2}{2} \leq x$, so y is bounded above by $y = \frac{x^2}{2}$. For $x \in [2, 3]$, $x \leq \frac{x^2}{2}$, so y is bounded above by $y = x$.

$$\begin{aligned} \text{Area} &= \int_0^2 \frac{x^2}{2} dx + \int_2^3 x dx \\ &= \left[\frac{x^3}{6} \right]_0^2 + \left[\frac{x^2}{2} \right]_2^3 \\ &= \frac{8}{6} + \left(\frac{9}{2} - \frac{4}{2} \right) \\ &= \frac{4}{3} + \frac{5}{2} \\ &= \frac{8+15}{6} = \frac{23}{6} \end{aligned}$$

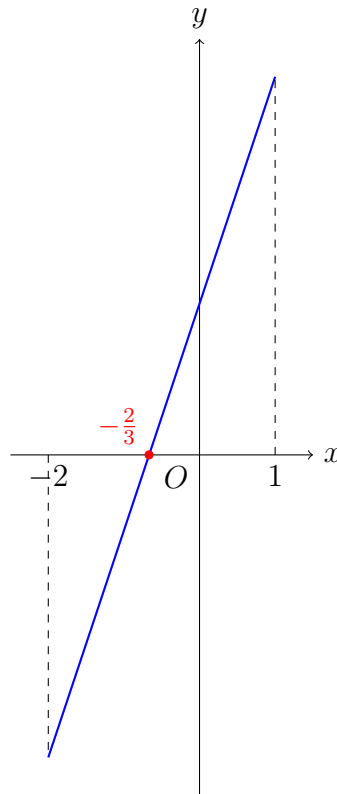
Final Answer

Area = $\frac{23}{6}$ sq. units

Question 34

Find the area of the region bounded by the line $y = 3x + 2$, the x -axis and the ordinates $x = -2$ and $x = 1$.

TYPE 5d Area of Piecewise-Defined or Sign-Changing Regions – Area with Sign Change



Region under $y = 3x + 2$ between $x = -2$ and $x = 1$

Given: Line $y = 3x + 2$, x -axis, and ordinates $x = -2, x = 1$

To Find: Total enclosed area

Concept / Formula Used: Area = $\int_{x_1}^{x_2} |y| dx$

Solution:

Find the x -intercept where $y = 0$:

$$3x + 2 = 0 \implies x = -\frac{2}{3}$$

For $x \in [-2, -\frac{2}{3}]$, $y \leq 0$, so $|y| = -(3x + 2)$. For $x \in [-\frac{2}{3}, 1]$, $y \geq 0$, so $|y| = 3x + 2$.

$$\begin{aligned} \text{Area} &= \int_{-2}^{-2/3} -(3x + 2) dx + \int_{-2/3}^1 (3x + 2) dx \\ &= \left[-\frac{3x^2}{2} - 2x \right]_{-2}^{-2/3} + \left[\frac{3x^2}{2} + 2x \right]_{-2/3}^1 \end{aligned}$$

Evaluating the first integral:

$$\begin{aligned} \left[-\frac{3x^2}{2} - 2x \right]_{-2}^{-2/3} &= \left(-\frac{3(4/9)}{2} - 2 \left(-\frac{2}{3} \right) \right) - \left(-\frac{3(4)}{2} - 2(-2) \right) \\ &= \left(-\frac{2}{3} + \frac{4}{3} \right) - (-6 + 4) \\ &= \frac{2}{3} + 2 = \frac{8}{3} \end{aligned}$$

Evaluating the second integral:

$$\begin{aligned} \left[\frac{3x^2}{2} + 2x \right]_{-2/3}^1 &= \left(\frac{3}{2} + 2 \right) - \left(\frac{3(4/9)}{2} + 2 \left(-\frac{2}{3} \right) \right) \\ &= \frac{7}{2} - \left(\frac{2}{3} - \frac{4}{3} \right) \\ &= \frac{7}{2} + \frac{2}{3} = \frac{25}{6} \end{aligned}$$

Total Area:

$$\text{Area} = \frac{8}{3} + \frac{25}{6} = \frac{16 + 25}{6} = \frac{41}{6}$$

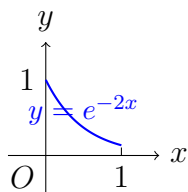
Final Answer

Area = $\frac{41}{6}$ sq. units

Question 35 (i)

Find the area under $y = e^{-2x}$ from $x = 0$ to $x = 1$.

TYPE 1f Area Under a Single Curve – Area Under Exponential Curve



Area under $y = e^{-2x}$ from $x = 0$ to $x = 1$

Given: Function $y = e^{-2x}$ on $[0, 1]$

To Find: Area under the curve

Concept / Formula Used: $\int e^{kx} dx = \frac{e^{kx}}{k} + c$

Solution:

Calculate the definite integral:

$$\begin{aligned}
 \text{Area} &= \int_0^1 e^{-2x} dx \\
 &= \left[\frac{e^{-2x}}{-2} \right]_0^1 \\
 &= -\frac{1}{2} (e^{-2} - e^0) \\
 &= -\frac{1}{2} (e^{-2} - 1) \\
 &= \frac{1}{2} (1 - e^{-2})
 \end{aligned}$$

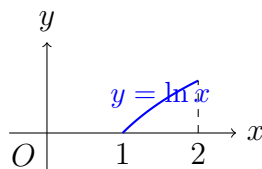
Final Answer

$$\text{Area} = \frac{1}{2} (1 - e^{-2}) \text{ sq. units}$$

Question 35 (ii)

Find the area under $y = \ln x$ from $x = 1$ to $x = 2$.

TYPE 1e Area Under a Single Curve – Area Under Logarithmic Curve



$$\text{Area under } y = \ln x \text{ from } x = 1 \text{ to } x = 2$$

Given: Function $y = \ln x$ on $[1, 2]$

To Find: Area under the curve

Concept / Formula Used: $\int \ln x dx = x \ln x - x + c$

Solution:

Apply integration by parts to evaluate $\int \ln x dx$:

$$\begin{aligned}
 \int \ln x dx &= \ln x \cdot x - \int \frac{1}{x} \cdot x dx \\
 &= x \ln x - x + c
 \end{aligned}$$

Evaluating the definite integral from 1 to 2:

$$\begin{aligned}
 \text{Area} &= [x \ln x - x]_1^2 \\
 &= (2 \ln 2 - 2) - (1 \ln 1 - 1) \\
 &= 2 \ln 2 - 2 - (0 - 1) \\
 &= 2 \ln 2 - 1
 \end{aligned}$$

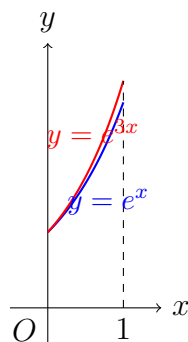
Final Answer

Area = $(2 \ln 2 - 1)$ sq. units

Question 35 (iii)

Find the area between $y = e^x$ and $y = e^{3x}$ from $x = 0$ to $x = 1$.

TYPE 2g Area Between Two Curves (Intersection-Bounded) – Area Between Exponential Curves



Area between $y = e^{3x}$ and $y = e^x$ on $[0, 1]$

Given: Curves $y = e^{3x}$ and $y = e^x$ on $[0, 1]$

To Find: Bounded area

Concept / Formula Used: Area = $\int_0^1 (e^{3x} - e^x) dx$

Solution:

For $x \in [0, 1]$, $e^{3x} \geq e^x$:

$$\begin{aligned}
 \text{Area} &= \int_0^1 (e^{3x} - e^x) dx \\
 &= \left[\frac{e^{3x}}{3} - e^x \right]_0^1 \\
 &= \left(\frac{e^3}{3} - e^1 \right) - \left(\frac{e^0}{3} - e^0 \right) \\
 &= \frac{e^3}{3} - e - \left(\frac{1}{3} - 1 \right) \\
 &= \frac{e^3}{3} - e + \frac{2}{3} \\
 &= \frac{1}{3} (e^3 - 3e + 2)
 \end{aligned}$$

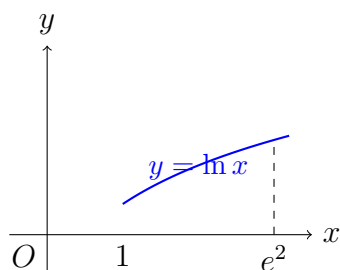
Final Answer

Area = $\frac{1}{3} (e^3 - 3e + 2)$ sq. units

Question 35 (iv)

Find the area between $y = \ln x$ and the x -axis from $x = 1$ to $x = e^2$.

TYPE 1e Area Under a Single Curve – Area Under Logarithmic Curve



Area under $y = \ln x$ from $x = 1$ to $x = e^2$

Given: Function $y = \ln x$ from $x = 1$ to $x = e^2$

To Find: Area under the curve

Concept / Formula Used: $\int \ln x dx = x \ln x - x + c$

Solution:

Integrating $\ln x$:

$$\begin{aligned}
 \text{Area} &= \int_1^{e^2} \ln x \, dx \\
 &= [x \ln x - x]_1^{e^2} \\
 &= (e^2 \ln(e^2) - e^2) - (1 \ln 1 - 1) \\
 &= (e^2 \cdot 2 - e^2) - (0 - 1) \\
 &= e^2 + 1
 \end{aligned}$$

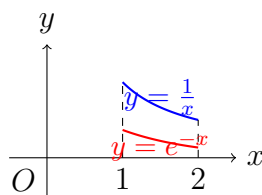
Final Answer

Area = $(e^2 + 1)$ sq. units

Question 35 (v)

Find the area between $y = e^{-x}$ and $y = \frac{1}{x}$ from $x = 1$ to $x = 2$.

TYPE 2a Area Between Two Curves (Intersection-Bounded) – Area Between Curves



Area between $y = \frac{1}{x}$ and $y = e^{-x}$ from $x = 1$ to $x = 2$

Given: Curves $y = \frac{1}{x}$ and $y = e^{-x}$ on $[1, 2]$

To Find: Area between the curves

Concept / Formula Used: Area = $\int_1^2 \left(\frac{1}{x} - e^{-x} \right) dx$

Solution:

For $x \in [1, 2]$, $\frac{1}{x} > e^{-x}$:

$$\begin{aligned}
 \text{Area} &= \int_1^2 \left(\frac{1}{x} - e^{-x} \right) dx \\
 &= [\ln |x| + e^{-x}]_1^2 \\
 &= (\ln 2 + e^{-2}) - (\ln 1 + e^{-1}) \\
 &= \ln 2 + e^{-2} - e^{-1} \\
 &= e^{-2} - e^{-1} + \ln 2
 \end{aligned}$$

Note: Written as positive magnitude $e^{-1} - e^{-2} - \ln 2$ in some textbooks depending on sign convention.

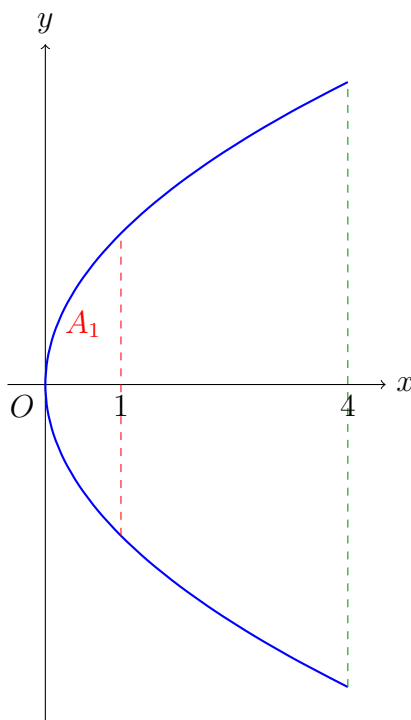
Final Answer

Area = $(e^{-2} - e^{-1} + \ln 2)$ sq. units

Question 36

If A_1 denotes the area of region bounded by $y^2 = 4x$, $x = 1$ and x -axis in the first quadrant and A_2 denotes the area of region bounded by $y^2 = 4x$, $x = 4$, find $A_1 : A_2$.

TYPE 2k Area Between Two Curves (Intersection-Bounded) – Ratio of Bounded Parabolic Areas



Regions A_1 (bounded by $x = 1$ in first quadrant) and total region up to $x = 4$

Given: Parabola $y^2 = 4x$, regions A_1 (first quadrant up to $x = 1$) and A_2 (entire region up to $x = 4$)

To Find: Ratio $A_1 : A_2$

Concept / Formula Used: Area = $\int_a^b y \, dx$

Solution:

Calculate A_1 , the area bounded by $y = 2\sqrt{x}$ from $x = 0$ to $x = 1$ in the first quadrant:

$$\begin{aligned} A_1 &= \int_0^1 2\sqrt{x} \, dx \\ &= 2 \left[\frac{2}{3} x^{3/2} \right]_0^1 \\ &= \frac{4}{3} \end{aligned}$$

Calculate A_2 , the full region bounded by $y^2 = 4x$ and $x = 4$ (including both top and bottom halves):

$$\begin{aligned} A_2 &= 2 \int_0^4 2\sqrt{x} \, dx \\ &= 4 \left[\frac{2}{3} x^{3/2} \right]_0^4 \\ &= \frac{8}{3} (4^{3/2}) \\ &= \frac{8}{3} (8) = \frac{64}{3} \end{aligned}$$

Now, compute the ratio $A_1 : A_2$:

$$\begin{aligned} \frac{A_1}{A_2} &= \frac{4/3}{64/3} \\ &= \frac{4}{64} \\ &= \frac{1}{16} \end{aligned}$$

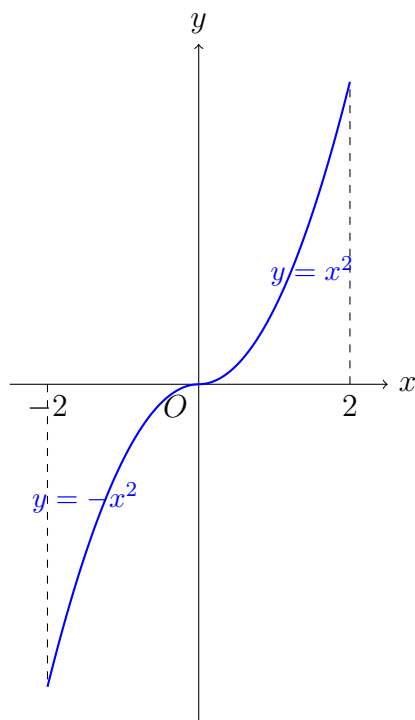
Final Answer

$$A_1 : A_2 = 1 : 16$$

Question 37

Sketch the graph of $y = x|x|$ and hence find the area bounded by this curve, x -axis and the ordinates $x = -2$ and $x = 2$ using integration.

TYPE 5b Area of Piecewise-Defined or Sign-Changing Regions – Area Bounded by Symmetric Piecewise Curve



Graph of $y = x|x|$ from $x = -2$ to $x = 2$

Given: Function $y = x|x|$ on $[-2, 2]$

To Find: Area bounded by the curve, x -axis, and $x = -2, x = 2$

Concept / Formula Used: $x|x| = \begin{cases} x^2, & x \geq 0 \\ -x^2, & x < 0 \end{cases}$

Solution:

Write $y = x|x|$ as a piecewise function:

$$y = \begin{cases} -x^2, & x < 0 \\ x^2, & x \geq 0 \end{cases}$$

The total area bounded by the curve and the x -axis is:

$$\begin{aligned} \text{Area} &= \int_{-2}^0 |-x^2| dx + \int_0^2 x^2 dx \\ &= \int_{-2}^0 x^2 dx + \int_0^2 x^2 dx \end{aligned}$$

Evaluating both integrals:

$$\begin{aligned} \int_{-2}^0 x^2 dx &= \left[\frac{x^3}{3} \right]_{-2}^0 = 0 - \left(-\frac{8}{3} \right) = \frac{8}{3} \\ \int_0^2 x^2 dx &= \left[\frac{x^3}{3} \right]_0^2 = \frac{8}{3} \end{aligned}$$

Total Area:

$$\text{Area} = \frac{8}{3} + \frac{8}{3} = \frac{16}{3}$$

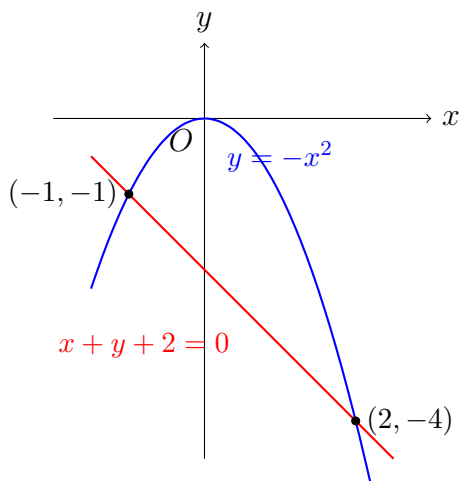
Final Answer

$$\text{Area} = \frac{16}{3} \text{ sq. units}$$

Question 38

Draw a rough sketch and find the area enclosed by the curve $y = -x^2$ and the line $x + y + 2 = 0$.

TYPE 2d Area Between Two Curves (Intersection-Bounded) – Area Between Parabola and Line



Region enclosed by parabola $y = -x^2$ and line $y = -x - 2$

Given: Parabola $y = -x^2$ and line $x + y + 2 = 0 \implies y = -x - 2$

To Find: Area enclosed between the curves

Concept / Formula Used: $\text{Area} = \int_{x_1}^{x_2} (y_{\text{upper}} - y_{\text{lower}}) dx$

Solution:

To find the intersection points, set $-x^2 = -x - 2$:

$$\begin{aligned} x^2 - x - 2 &= 0 \\ (x - 2)(x + 1) &= 0 \implies x = -1 \text{ or } x = 2 \end{aligned}$$

For $x \in [-1, 2]$, $y = -x^2$ lies above $y = -x - 2$:

$$\begin{aligned}\text{Area} &= \int_{-1}^2 (-x^2 - (-x - 2)) dx \\ &= \int_{-1}^2 (2 + x - x^2) dx \\ &= \left[2x + \frac{x^2}{2} - \frac{x^3}{3} \right]_{-1}^2\end{aligned}$$

Evaluating at upper limit $x = 2$:

$$2(2) + \frac{2^2}{2} - \frac{2^3}{3} = 4 + 2 - \frac{8}{3} = \frac{10}{3}$$

Evaluating at lower limit $x = -1$:

$$2(-1) + \frac{(-1)^2}{2} - \frac{(-1)^3}{3} = -2 + \frac{1}{2} + \frac{1}{3} = -\frac{7}{6}$$

Subtracting:

$$\begin{aligned}\text{Area} &= \frac{10}{3} - \left(-\frac{7}{6} \right) \\ &= \frac{20}{6} + \frac{7}{6} \\ &= \frac{27}{6} = \frac{9}{2}\end{aligned}$$

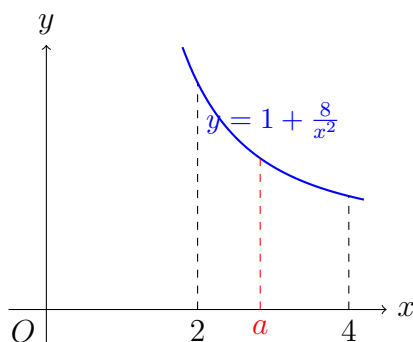
Final Answer

Area = $\frac{9}{2}$ sq. units

Question 39

Find the area of the region bounded by the x -axis, part of the curve $y = 1 + \frac{8}{x^2}$, and the ordinates $x = 2$ and $x = 4$. If the ordinate at $x = a$ divides the area into two equal parts, find the value of a .

TYPE 5c Area of Piecewise-Defined or Sign-Changing Regions – Area Division by Line $x=a$



Region under $y = 1 + \frac{8}{x^2}$ from $x = 2$ to $x = 4$ divided at $x = a$

Given: Curve $y = 1 + \frac{8}{x^2}$, ordinates $x = 2, x = 4$, divided equally by $x = a$

To Find: Value of a

Concept / Formula Used: $\int_2^a y \, dx = \frac{1}{2} \int_2^4 y \, dx$

Solution:

First, compute the total area from $x = 2$ to $x = 4$:

$$\begin{aligned} \text{Total Area} &= \int_2^4 \left(1 + \frac{8}{x^2}\right) dx \\ &= \left[x - \frac{8}{x}\right]_2^4 \\ &= \left(4 - \frac{8}{4}\right) - \left(2 - \frac{8}{2}\right) \\ &= (4 - 2) - (2 - 4) \\ &= 2 - (-2) = 4 \end{aligned}$$

Since $x = a$ divides the region into two equal parts, the area from $x = 2$ to $x = a$ must equal 2:

$$\begin{aligned} \int_2^a \left(1 + \frac{8}{x^2}\right) dx &= 2 \\ \left[x - \frac{8}{x}\right]_2^a &= 2 \\ \left(a - \frac{8}{a}\right) - \left(2 - \frac{8}{2}\right) &= 2 \\ a - \frac{8}{a} - (-2) &= 2 \\ a - \frac{8}{a} + 2 &= 2 \\ a - \frac{8}{a} &= 0 \\ a^2 &= 8 \end{aligned}$$

Since $2 < a < 4$:

$$a = \sqrt{8} = 2\sqrt{2}$$

Final Answer

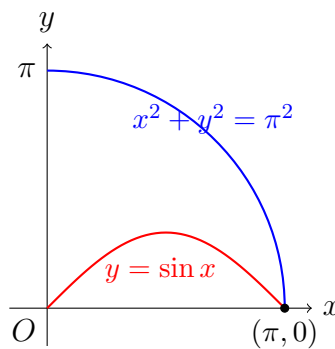
$$a = 2\sqrt{2}$$

Question 40

There are two curves given in the first quadrant as: $x^2 + y^2 = \pi^2$, $y = \sin x$.

- What are the points of intersection of both the given curves?
- What is the value of k , if $\int_0^\pi \sqrt{\pi^2 - x^2} dx = \frac{\pi^3}{k}$?
- Sketch the region enclosed by the given curves in the first quadrant and the y -axis.
- Find the area of the region enclosed by the given curves in the first quadrant and the y -axis.

TYPE 2h Area Between Two Curves (Intersection-Bounded) – Area Between Circle and Trigonometric Function



Region between $x^2 + y^2 = \pi^2$ and $y = \sin x$ in first quadrant

Given: Curves $x^2 + y^2 = \pi^2$ and $y = \sin x$ in the first quadrant

To Find: Intersection points, constant k , rough sketch, and bounded area

Concept / Formula Used: $\int_0^\pi \sqrt{\pi^2 - x^2} dx = \frac{\pi^3}{4}$ $\int \sin x dx = -\cos x$

Solution:

(a) **Point of intersection:** On the x -axis at $x = \pi$: For circle: $y = \sqrt{\pi^2 - \pi^2} = 0$. For sine wave: $y = \sin \pi = 0$. For $x \in [0, \pi)$, $\sqrt{\pi^2 - x^2} > 1 \geq \sin x$, so there are no other points of intersection. The point of intersection in the first quadrant is $(\pi, 0)$.

(b) **Finding k :** The integral represents the area of a quarter circle of radius π :

$$\int_0^\pi \sqrt{\pi^2 - x^2} dx = \frac{1}{4}\pi(\pi)^2 = \frac{\pi^3}{4}$$

Equating to $\frac{\pi^3}{k}$:

$$\frac{\pi^3}{4} = \frac{\pi^3}{k} \implies k = 4$$

(c) **Sketch of the region:** The region is bounded by $y = \sqrt{\pi^2 - x^2}$ from above, $y = \sin x$ from below, the y -axis ($x = 0$) on the left, up to $x = \pi$.

(d) Area calculation:

$$\begin{aligned}\text{Area} &= \int_0^{\pi} \left(\sqrt{\pi^2 - x^2} - \sin x \right) dx \\&= \int_0^{\pi} \sqrt{\pi^2 - x^2} dx - \int_0^{\pi} \sin x dx \\&= \frac{\pi^3}{4} - [-\cos x]_0^{\pi} \\&= \frac{\pi^3}{4} - (-\cos \pi - (-\cos 0)) \\&= \frac{\pi^3}{4} - (1 + 1) \\&= \frac{\pi^3}{4} - 2\end{aligned}$$

Final Answer

- (a) Intersection point: $(\pi, 0)$
- (b) $k = 4$
- (c) As sketched above
- (d) Area = $\left(\frac{\pi^3}{4} - 2\right)$ sq. units